A Data Construction

A.1 Disaster data and episode construction

TABLE A.1: DISASTER SAMPLES: HAZARD COMPOSITION AND EPISODE CHARACTERISTICS

Panel A. Broad disaster sample, 2009–2024

Hazard	Episodes	Share (%)	Mean damage ($ millions)	Mean fatalities	Mean injuries	Median affected counties
Flood	240	21.8	606.2	3.31	2.82	5.5
Heat	233	21.2	0.0	14.03	33.09	1.0
Tornado	129	11.7	260.1	8.26	81.91	7.0
Winter weather	109	9.9	89.2	4.27	34.61	7.0
Hail	89	8.1	324.2	0.11	1.60	3.0
Drought	75	6.8	268.0	0.00	4.67	5.0
Wildfire	61	5.6	613.8	4.61	6.44	2.0
Hurricane	57	5.2	1,799.3	8.39	4.49	8.0
Severe storm	38	3.5	95.3	2.26	27.16	5.5
Wind	29	2.6	268.5	3.83	6.48	12.0
Coastal	24	2.2	1,539.3	4.04	0.54	5.5
Other	15	1.4	75.7	8.07	31.00	2.0
All episodes	1,099	100.0	388.7	6.17	23.26	4.0

Panel B. Selected flood episodes, 2009–2024

Characteristic	Episodes	Mean	Std. dev.	P25	Median	P75	P90
Damage ($ millions)	1,490	101.59	1,546.87	1.57	3.36	10.74	46.75
Fatalities	1,490	0.85	3.41	0.00	0.00	0.00	2.00
Injuries	1,490	0.74	5.67	0.00	0.00	0.00	1.00
Affected counties	1,490	4.82	5.44	1.00	3.00	6.00	10.10
Duration (days)	1,490	4.49	6.52	1.00	2.00	4.00	11.00

Panel C. Selected flood, hurricane, wildfire, and coastal episodes, 2009–2024

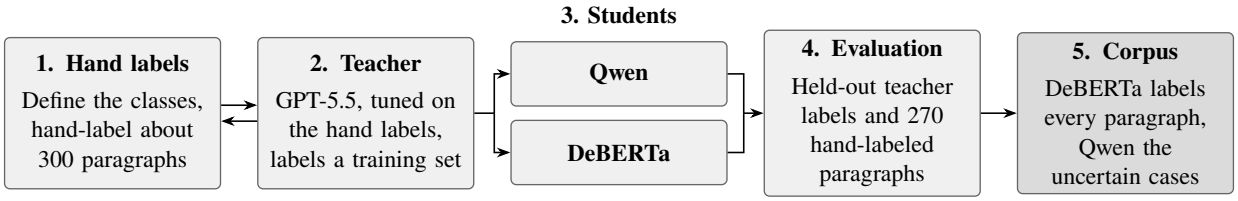
Characteristic	Episodes	Mean	Std. dev.	P25	Median	P75	P90
Damage ($ millions)	1,938	167.11	1,776.55	1.33	3.37	12.26	61.38
Fatalities	1,938	1.27	5.23	0.00	0.00	1.00	3.00
Injuries	1,938	1.38	6.68	0.00	0.00	0.00	3.00
Affected counties	1,938	4.50	5.56	1.00	3.00	6.00	10.00
Duration (days)	1,938	5.04	7.13	1.00	2.00	5.00	15.00

Notes. Panel A describes the broad disaster sample used in. Panel B describes the flood episodes used in the main flood analysis. Panel C describes the flood, hurricane, wildfire, and coastal episodes used in the main analysis. Damage is expressed in 2023 dollars. Fatalities, injuries, and affected counties are episode-level measures; the affected-counties measure counts distinct U.S. counties linked to an episode. Other combines hazard categories represented by fewer than ten episodes.

A.2 Local news corpus

A.3 The interpretation classifier

FIGURE A.1: THE CLASSIFICATION PIPELINE



Notes. Steps of the classification of disaster articles at the paragraph level. I define the classes and hand-label about 300 paragraphs, which I use to tune the prompts of the teacher model, GPT-5.5. The teacher labels a training set, on which I train two local student models, DeBERTa and Qwen. I evaluate the students on held-out teacher labels and on 270 hand-labeled paragraphs (Appendix Tables A.2 and A.3). DeBERTa classifies the full corpus and passes the uncertain paragraphs to Qwen.

TABLE A.2: CANDIDATE MODELS AGAINST THE REFERENCE LABELS

Model	disaster (1)	specific (2)	link (3)	skeptical (4)	human (5)	flip↓ (6)
Qwen3.5-9B-staged	.952	.960	.947	.911	.962	.015
DeBERTa-v3-large	.905	.932	.938	.880	.909	.015
gpt-5.4-mini	.877	.917	.906	.818	.954	.053
gpt-4.1-mini	.893	.895	.899	.774	.893	.075
gpt-4o-mini	.745	.931	.739	.804	.911	.063

Notes. Positive-class F_1 score of each candidate model against the reference labels, on a random sample of blocks. Flip is the share of blocks whose label the model reverses.

TABLE A.3: CANDIDATE MODELS AGAINST HAND LABELS ON THE HELD-OUT BLOCKS

Model	disaster (1)	link (2)	skeptical (3)	flip↓ (4)
Qwen3.5-9B-staged	.888	.876	.918	.009
DeBERTa-v3-large	.876	.868	.871	.027
gpt-5.4-mini	.804	.768	.830	.029
gpt-4.1-mini	.869	.846	.800	.059
gpt-4o-mini	.667	.638	.844	.077

Notes. Positive-class F_1 score of each candidate model against my hand labels on the held-out blocks, the hardest cases of the labeling set. See notes to Table A.2.

TABLE A.4: EXAMPLE PARAGRAPHS AND THEIR CLASSIFICATION

Climate link Hurricane Helene, Associated Press, October 9, 2024	**Human toll and personal story** Hurricane Katrina, *Wall Street Journal*, August 31, 2005
"The scientists warned that **continued burning of fossil fuels will lead to more hurricanes like Helene**, with 'unimaginable' floods well inland, not just on coasts." *Classification.* Specific disaster event; affirming climate link, central to the paragraph. The paragraph never mentions "climate change."	"After climbing out of one boat, Judy Martin took a few steps and then **collapsed onto the highway** [. . .] Ms. Martin said that **her 95-year-old mother had drowned in their house** as the floodwaters rose around them." *Classification.* Specific disaster event; human toll, told as one person's story; no climate link.

Notes. Two paragraphs from the coverage of disaster events and the labels the classifier assigns to them. Bold text marks the passages that determine the label.

A.4 Topic hierarchy and composition

A.5 Circulation and county exposure weights

A.6 The interpretation measure

TABLE A.5: AGREEMENT BETWEEN THE COUNTY NEWS CLASSES AND PUBLISHED NATIONAL CLIMATE-NEWS INDICES

	EGKLS WSJ (monthly)		MCCC (weekly)		NYT (monthly)	
News class	Level (1)	Innov. (2)	Level (3)	Innov. (4)	Level (5)	Innov. (6)
Panel A. Climate-topic classes						
Climate, no disaster, plus climate policy and green energy	.565	.333	−.011	.313	.159	.126
Climate coverage, no disaster	.564	.411	.227	.291	.305	.202
Climate policy and green energy	.550	.282	−.027	.309	.142	.106
Climate or peripheral (disaster-contaminated)	.318	.253	−.462	.299	−.580	.064
Panel B. Disaster classes						
Disaster coverage with a climate attribution	.475	.333	.647	.278	.709	.237
All disaster coverage	.088	.037	−.516	.019	−.553	−.059
Disaster policy, preparedness, insurance	−.067	.020	−.592	.046	−.683	−.132
Panel C. Pooled series, denominator and placebo						
All scored coverage except irrelevant	.207	.099	−.534	.149	−.583	−.034
All scored coverage	.140	.070	−.571	.127	−.628	−.045
Irrelevant coverage (placebo)	−.086	−.054	−.633	−.002	−.709	−.098
Periods (levels / innovations)	210 / 209		1,074 / 916		288 / 287	

Notes. This table reports the agreement between the county news classes and three published national climate-news indices. Each class is aggregated to a county panel, standardized within county, and reduced to a national common factor by principal components across counties. "Level" is the correlation of that first principal component with the published index in levels; "Innov." is the correlation of their innovations. Innovations are AR(1) forecast errors from the prior 156 weeks for the weekly series; for MCCC and EGKLS they are the innovation series the authors themselves publish.

The frequencies are not the same across columns because the published series are not. EGKLS is monthly and ends in June 2017; MCCC is daily, aggregated here to weeks; the NYT index is daily.

The placebo row is coverage the classifier marks as unrelated to climate.

A.7 Adviser identity, geography, and screens

A.8 13F holdings, identifier matching, and market data

A.9 Green classification of ETFs

A.10 Green classification of mutual funds

A.11 Newspaper ownership changes

A.12 The green ETF segment

A.13 Mutual funds: S12 accounting and manager geography

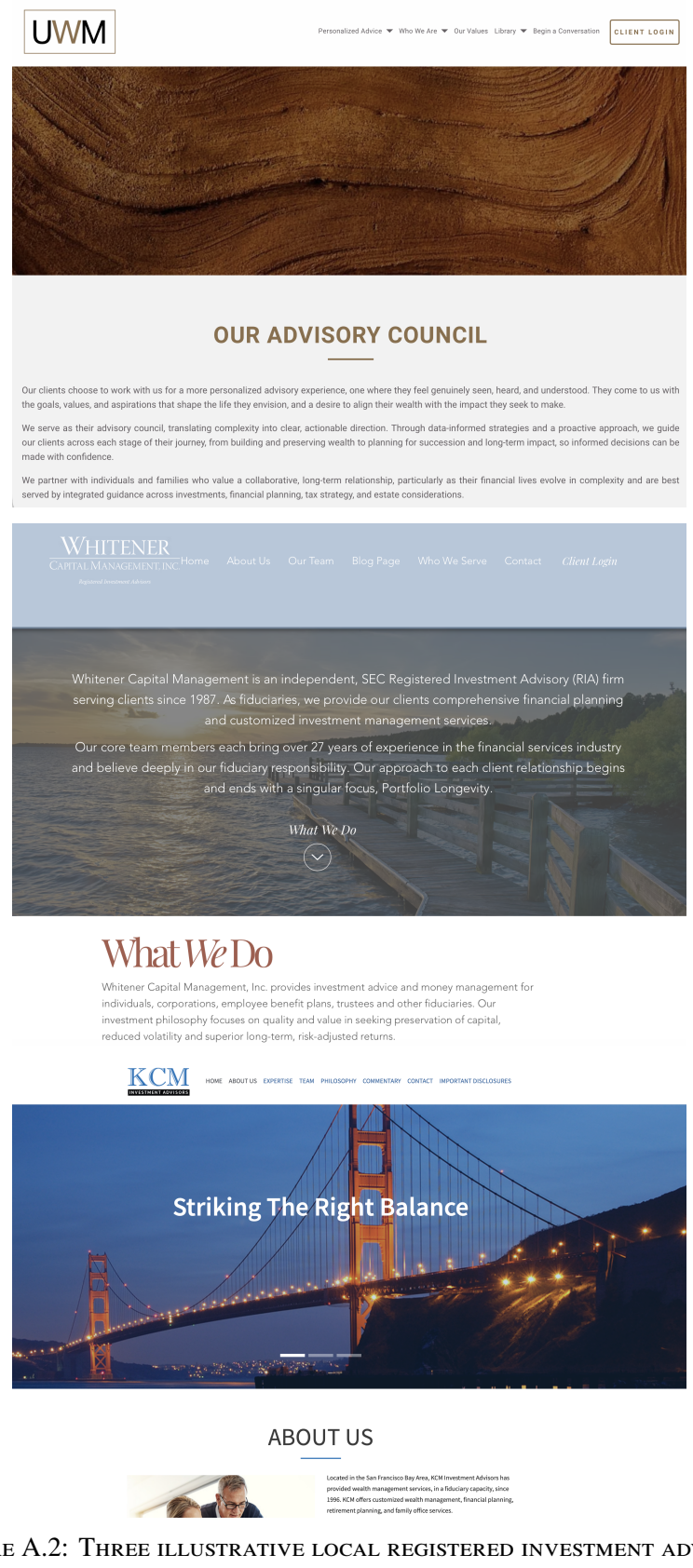


FIGURE A.2: THREE ILLUSTRATIVE LOCAL REGISTERED INVESTMENT ADVISERS

TABLE A.6: BUSINESS-TYPE COMPOSITION OF REGISTERED INVESTMENT ADVISERS, 2009–2024

Business type	All registered advisers (%)	Household-serving advisers (%)
Investment adviser	55.5	66.4
Hedge fund manager	20.4	1.6
Private banking/wealth management	19.1	30.6
Venture capital/private equity	2.4	0.3
Other	2.6	1.1

Notes. This table reports the business-type composition of registered investment advisers over 2009–2024. Entries are percentages of advisers, classified using each adviser's latest observation. Advisers serving predominantly household accounts have a strict majority of individual and high-net-worth individual clients. Business types representing less than 1 percent of all registered advisers are combined as Other; the same grouping is used in both columns.

FIGURE A.3: MARKET VALUE OF THE GREEN SEGMENT AND OF THE ETF UNIVERSE, 2005–2024

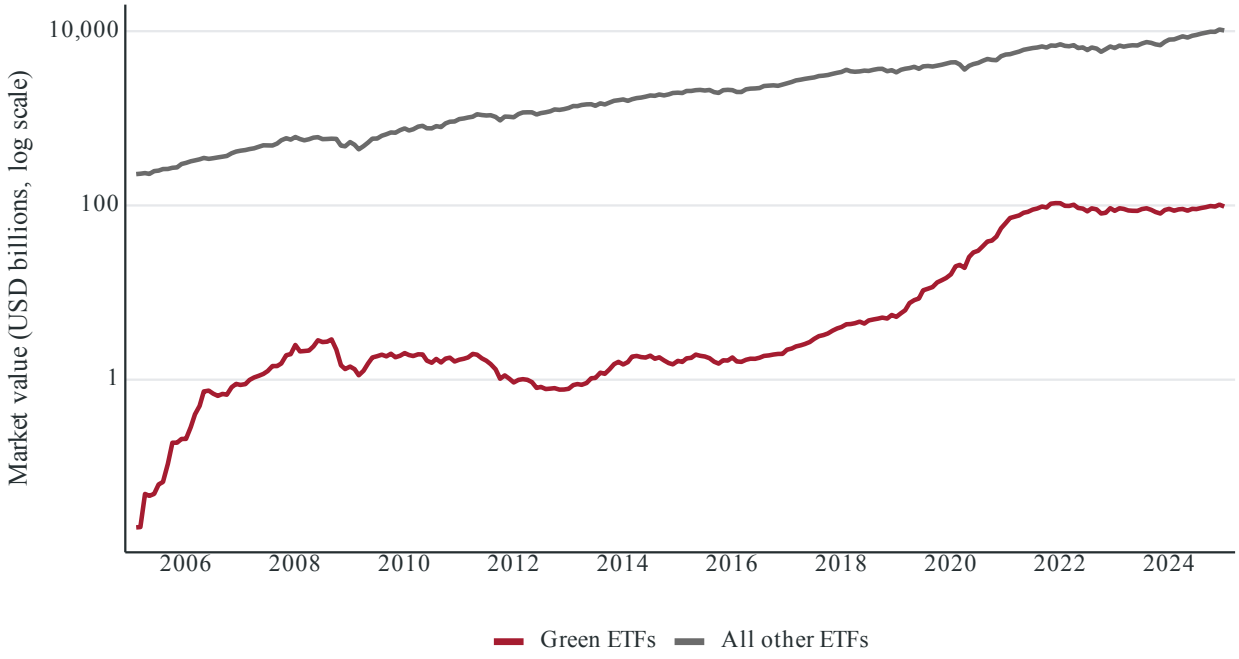
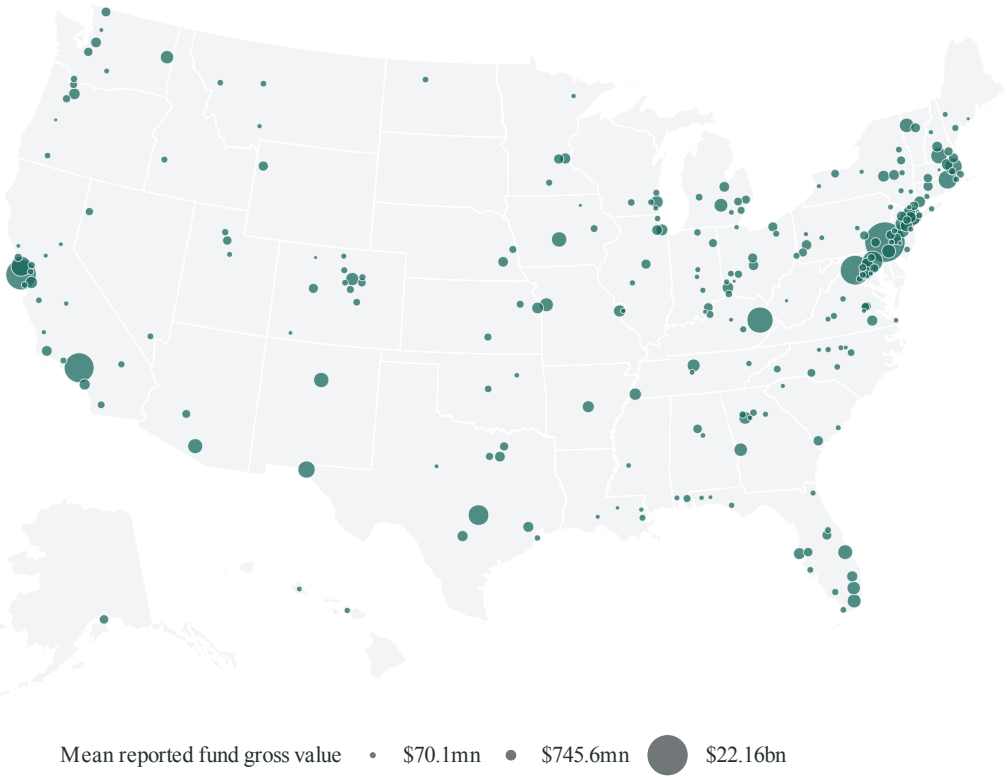


Figure A.4: Mutual-fund management geography, 2009–2024. Funds whose managers or managing firms span several counties are assigned proportional county weights.



A.14 Auxiliary datasets

B Conceptual Framework and Empirical Strategy

I use a portfolio-choice model to describe how interpreting a disaster as evidence of climate change affects demand for ESG funds. The model links the local allocation response to investors' preferences and belief updating. It then provides the additional allocation for the exercise in Appendix J, holding attention and the wider information environment fixed.

B.1 Portfolio choice and beliefs

Consider an investor choosing the share $q \in [0,1]$ of eligible wealth invested in a representative ESG portfolio. The remaining share, $1-q$, is invested in an other-fund portfolio. Let $P > 0$ be the price of the ESG portfolio relative to the other-fund portfolio, whose price is normalized to one. Initial eligible capital is normalized to one. If q_0 is the inherited ESG share at relative price P^0, wealth before rebalancing is $w(P) = q_0 P/P^0 + 1 - q_0$. The chosen shares purchase $qw(P)/P$ units of the ESG portfolio and $(1-q)w(P)$ units of the other portfolio, so the budget is satisfied without external deposits or withdrawals.

There are two future climate states, adverse h and benign l. In state k, the ESG and other portfolios pay $X_{Ek} > 0$ and $X_{Ok} > 0$ per unit, respectively. The investor incurs background loss L_k and receives a nonfinancial benefit ψg_k per ESG unit in consumption units: ψ measures preference strength and g_k the state-specific benefit. Let

$$r_k(P) = \frac{X_{Ek} + \psi g_k}{P} - X_{Ok}, \qquad k \in \{h, l\}, \tag{B.1}$$

be the payoff, including the holdings benefit, from shifting one dollar from the other-fund portfolio to the ESG portfolio. I assume $r_h(P) > 0 > r_l(P)$ over the relevant price range. The ESG tilt provides insurance in the adverse climate state at a cost in the benign state.

The other-fund portfolio costs one, so X_{Ok} is its return per dollar in state k, the benchmark return. The ESG portfolio costs P, so $(X_{Ek} + \psi g_k)/P$ is its return per dollar, the economic payoff plus the nonfinancial benefit, and $r_k(P)$ in equation (B.1) is the difference between the two. For perceived probability π of the adverse state, the investor has exponential utility of end-of-period consumption with risk aversion $\rho > 0$ and solves

$$\begin{aligned} \max_{0 \le q \le 1} \quad & -\pi \exp\left\{-\rho \left[w(P)\left((1-q)X_{Oh} + q\,\frac{X_{Eh} + \psi g_h}{P}\right) - L_h\right]\right\} \\ & -(1-\pi)\exp\left\{-\rho \left[w(P)\left((1-q)X_{Ol} + q\,\frac{X_{El} + \psi g_l}{P}\right) - L_l\right]\right\}, \end{aligned} \tag{B.2}$$

which, collecting the terms in q and using $r_k(P)$, is

$$\begin{aligned} \max_{0 \le q \le 1} \quad & -\pi \exp\{-\rho\,[w(P)X_{Oh} - L_h + w(P)q r_h(P)]\} \\ & -(1-\pi)\exp\{-\rho\,[w(P)X_{Ol} - L_l + w(P)q r_l(P)]\}. \end{aligned} \tag{B.3}$$

The news changes the perceived probability of each climate state, while asset payoffs and losses within each state, as well as investors' preferences, remain unchanged. The following derivation considers interior portfolios.

For episode e, let $f_i(z_e \mid k)$ be investor i's subjective likelihood of disaster news z_e in state k. Define the likelihood ratio $\mathrm{LR}_{i,e} = f_i(z_e \mid h)/f_i(z_e \mid l)$ and its logarithm $\ell_{i,e} = \log \mathrm{LR}_{i,e}$.

The event indicator η_e is zero before disaster news and one afterward. Let $\pi_{0i} \in (0,1)$ and π_{1i} be the

prior and posterior probabilities of the adverse state, with odds $\omega_{0i} = \pi_{0i}/(1-\pi_{0i})$ and $\omega_{1i} = \pi_{1i}/(1-\pi_{1i})$. Bayes' rule gives

$$\omega_{1i} = \omega_{0i}\mathrm{LR}_{i,e}^{\eta_e}, \qquad \log\omega_{1i} = \log\omega_{0i} + \eta_e \ell_{i,e}. \tag{B.4}$$

Positive $\ell_{i,e}$ raises perceived adverse-climate risk. Interpretation may reflect investors' own models or supplied narratives. The benchmark shares preferences, prior odds $\omega_{0i} = \omega_0$, and initial normalized portfolios; investors do not infer additional information from prices.

B.2 The predicted portfolio response

Dividing expected utility by $1-\pi$ leaves the optimal portfolio unchanged. Beliefs then enter through the odds of the adverse state relative to the benign state, $\omega = \pi/(1-\pi)$. Differentiating with respect to q and dividing by $\rho w(P)$ gives the optimality condition

$$\begin{aligned} &\omega r_h(P) e^{-\rho[w(P)X_{Oh} - L_h + w(P)q r_h(P)]} \\ &\quad = [-r_l(P)] e^{-\rho[w(P)X_{Ol} - L_l + w(P)q r_l(P)]}. \end{aligned} \tag{B.5}$$

Denote the optimal ESG share by $q^*(P,\omega)$. Taking logarithms gives a linear equation in this share. Because news changes log odds by $\eta_e \ell_{i,e}$, the solution implies

$$q^*(P,\omega_{1i}) - q^*(P,\omega_0) = \eta_e \chi(P)\ell_{i,e}, \qquad \chi(P) = \frac{1}{\rho w(P)[r_h(P) - r_l(P)]} > 0. \tag{B.6}$$

Here $\chi(P)$ is the change in the ESG share per unit increase in log odds, holding the price fixed. The formula applies to finite belief updates on the interior support.

Let $q_0 = q^*(P^0,\omega_0)$ be the initial optimal share. At initial prices, the event changes desired allocation by

$$\begin{aligned} u_{i,e} &\equiv q^*(P^0,\omega_{1i}) - q_0 = \eta_e \chi(P^0)\ell_{i,e} = \eta_e \gamma_{i,e}, \\ \gamma_{i,e} &\equiv \chi(P^0)\ell_{i,e}. \end{aligned} \tag{B.7}$$

Thus $u_{i,e} = 0$ before news ($\eta_e = 0$) and $u_{i,e} = \gamma_{i,e}$ after news ($\eta_e = 1$). The event response $\gamma_{i,e}$ combines portfolio responsiveness and the belief update at initial prices.

At post-news relative price P_e, the observed share change is $\Delta q_{i,e} = q^*(P_e,\omega_{1i}) - q_0$. Adding and subtracting the post-news allocation at initial prices gives

$$\Delta q_{i,e} = u_{i,e} + \underbrace{q^*(P_e,\omega_{1i}) - q^*(P^0,\omega_{1i})}_{\text{response to the price change after news}}. \tag{B.8}$$

The ratio P_e/P^0 is the gross relative-price return, and $(P_e - P^0)/P^0$ is the corresponding net return. Thus $\Delta p_e = \log(P_e/P^0) = \log(1 + (P_e - P^0)/P^0) \approx (P_e - P^0)/P^0$. No relative-price change means gross return one, net return zero, and $\Delta p_e = 0$. The expansion around $\Delta p_e = 0$ therefore concerns small relative-price returns. Holding updated beliefs fixed gives

$$\begin{aligned} \Delta q_{i,e} &= u_{i,e} - \zeta_{i,e}^q \Delta p_e + o(|\Delta p_e|), \\ \zeta_{i,e}^q &= -\left.\frac{\partial q^*(P,\omega_{1i})}{\partial \log P}\right|_{P=P^0}. \end{aligned} \tag{B.9}$$

The coefficient $\zeta^q_{i,e}$ measures the response of the portfolio share to prices. Its independence from the news is an additional local restriction.

Assumption: price responsiveness. Disaster exposure and stronger climate interpretation shift desired allocations but do not change investors' responsiveness to prices. Conditional on the controls, average price responsiveness does not systematically differ with disaster exposure or interpretation.[60]

I impose this assumption on the local price-response term in (B.9). The common price movement then contributes an episode component to the conditional mean, while differences associated with disaster exposure and interpretation reflect $u_{i,e}$. For holdings valued at initial prices, I impose the corresponding conditional-average restriction on the price response of that measured share.[61]

B.3 From the model to the estimating equation

Let $A_{i,e}$ denote attention to disaster news, $I_{i,e}$ its climate interpretation, and Z_e the wider information environment. Write the belief update as $\ell_{i,e} = \ell_i(A_{i,e}, I_{i,e}, Z_e)$ and the allocation response as $u_{i,e}(A, I, Z) = \eta_e \chi(P^0)\ell_i(A, I, Z)$.

For county c, the impact outcome $\Delta q_{c,e} = q_{c,0} - q_{c,-1}$ is the ESG-share change from the pre-event quarter -1 to the event quarter 0. Let $I_{c,e}$ be the circulation-weighted average of outlets' shares of disaster articles linking the event to climate change. Write $I^*_{c,e} = (I_{c,e} - \bar{I})/\sigma_I$, where $\bar{I}$ and σ_I are its sample mean and standard deviation. Let $A_{c,e}$ be local disaster-news coverage and $A^*_{c,e}$ its standardized $\log(1 + A_{c,e})$. Set $H_{c,e} = 1$ for affected counties and zero for clean comparison counties in the same episode. All these counties are observed around a disaster, so $\eta_e = 1$.

Let $\ell_{c,e}$ be the conditional average belief update and $\ell^0_{c,e}$ its value at mean interpretation. Holding attention and controls fixed, I assume a unit increase in I^* changes the belief update by $\gamma^\ell_{1,e}$ among affected investors and $\gamma^\ell_{0,e}$ among comparison investors:

$$\ell_{c,e} = \ell^0_{c,e} + \gamma^\ell_{H_{c,e},e} I^*_{c,e}. \tag{B.10}$$

In the common-response benchmark, $u_{c,e} = \chi(P^0)\ell_{c,e}$. Write $\overline{\Delta q}_{c,e}$ for the predicted share change conditional on exposure, interpretation, attention, controls, and episode. Under the price-response assumption, comparing counties c and c' at the same controls within an episode gives

$$\overline{\Delta q}_{c,e} - \overline{\Delta q}_{c',e} \simeq u_{c,e} - u_{c',e} = \chi(P^0)\big(\ell_{c,e} - \ell_{c',e}\big). \tag{B.11}$$

The approximation inherits the price remainder in (B.9). Individual price responses may differ; the assumption concerns their conditional average in the comparison.

Let $b_e = -\bar{\zeta}^q_e \Delta p_e$ denote the common conditional price contribution, where $\bar{\zeta}^q_e$ is average price respon-

[60] A related benchmark in van der Beck et al. [2025], Assumption 1, allows heterogeneous price elasticities that are independent of the process generating demand shifts across investors.

[61] Let $q_i = q^*(P, \omega_{1i})$. The share of these holdings valued at P^0 is $q^{\mathrm{FP}}_i = (P^0/P)q_i/[1 - q_i + (P^0/P)q_i]$. At $P = P^0$, it equals q_i, so its initial-price information shift is also $u_{i,e}$. Its price response is obtained by expanding this measured share; its coefficient generally differs from $\zeta^q_{i,e}$.

siveness. Substituting the interpretation rule into the allocation response gives

$$\begin{aligned}\overline{\Delta q}_{c,e} \simeq b_e + \chi(P^0)\ell^0_{c,e} + \chi(P^0)\gamma^\ell_{0,e} I^*_{c,e} \\ + \chi(P^0)(\gamma^\ell_{1,e} - \gamma^\ell_{0,e}) H_{c,e} I^*_{c,e}.\end{aligned} \tag{B.12}$$

Assume a linear baseline response. Let λ_e denote episode effects, δ_H the exposure coefficient, β_A the comparison-county attention slope, and δ_A its affected-county increment. Remaining controls and other fixed effects $\mathbf{W}_{c,e}$ have coefficients θ:

$$\begin{aligned}b_e + \chi(P^0)\ell^0_{c,e} = \lambda_e + \delta_H H_{c,e} + \beta_A A^*_{c,e} \\ + \delta_A H_{c,e} A^*_{c,e} + \mathbf{W}'_{c,e}\theta.\end{aligned} \tag{B.13}$$

Episode effects absorb b_e; portfolio responsiveness $\chi(P^0)$ remains in the interpretation response.

With interpretation slope β_I in comparison counties, affected-county increment δ_I, and regression residual $\varepsilon_{c,e}$, the estimating equation is

$$\begin{aligned}\Delta q_{c,e} = \lambda_e + \delta_H H_{c,e} + \beta_A A^*_{c,e} + \beta_I I^*_{c,e} \\ + \delta_A H_{c,e} A^*_{c,e} + \delta_I H_{c,e} I^*_{c,e} + \mathbf{W}'_{c,e}\theta + \varepsilon_{c,e}.\end{aligned} \tag{B.14}$$

The residual includes individual deviations from the conditional price response and has zero conditional mean in the maintained first-order representation. Matching the interpretation terms gives

$$\beta_I = \chi(P^0)\gamma^\ell_{0,e}, \qquad \delta_I = \chi(P^0)(\gamma^\ell_{1,e} - \gamma^\ell_{0,e}). \tag{B.15}$$

This coefficient mapping assumes that the response products are common across episodes. Otherwise, the pooled coefficients summarize the heterogeneous products using the county–episode variation remaining after controls and fixed effects; they are not generally arithmetic averages.

The mapping therefore uses the portfolio model, the local price approximation and price-response assumption, the linear interpretation and baseline relations, and common response products across episodes. Under these assumptions, the interpretation effect among affected investors is

$$\beta_I + \delta_I = \underbrace{\chi(P^0)}_{\substack{\text{portfolio response} \\ \text{to a belief update}}} \underbrace{\gamma^\ell_{1,e}}_{\substack{\text{belief response} \\ \text{to interpretation}}} = \frac{\gamma^\ell_{1,e}}{\rho[r_h(P^0) - r_l(P^0)]}. \tag{B.16}$$

It measures the additional desired ESG share at initial prices from one standard deviation more climate interpretation, holding attention and controls fixed. Here $w(P^0) = 1$ by the initial wealth normalization.

For stronger interpretation I^1 relative to baseline I^0 in the same county and event, define $a^I_{i,e} = u_{i,e}(A, I^1, Z) - u_{i,e}(A, I^0, Z)$. Appendix J aggregates its estimated counterpart,

$$\widehat{a}^I = (\widehat{\beta}_I + \widehat{\delta}_I)\frac{I^1 - I^0}{\sigma_I}. \tag{B.17}$$

B.4 Clean windows, the choice of estimator, and weighting

B.5 The within-episode design

C Appendix to Section 2

TABLE C.1: SCALE AND HOUSEHOLD RELEVANCE OF THE ADVISER ETF CHANNEL

	All registered advisers	Household-serving advisers
Panel A. ETF adoption among registered investment advisers, 2009–2024		
Registered investment advisers	8,489	4,439
Adviser-quarters holding any ETF (%)	72.1	87.4
Advisers ever holding any ETF (%)	87.1	97.3
Adviser-quarters holding a green ETF (%)	16.4	19.8
Advisers ever holding a green ETF (%)	36.3	45.6
	All single-county advisers	Household-serving single-county advisers
Panel B. Portfolio allocation among single-county ETF holders, 2009–2024		
ETF value / observed 13F value (%)	8.1	17.7
ETF value / (ETF + directly held stocks) (%)	8.4	18.3
Median adviser-quarter ETF value / observed 13F value (%)	19.1	28.1
Adviser-quarters with ETF value above directly held stocks (%)	34.8	39.6
Green-ETF value / ETF value (%)	0.9	1.1
Green-ETF value / observed 13F value (%)	0.07	0.19
	Share of total market (%)	Share of observed Form 13F value (%)
Panel C. Share of the U.S. ETF market by reporting sector, 2024Q4 ($10,294.1 billion)		
All registered RIAs	41.1	67.0
Household-serving single-county RIAs	8.1	13.1
Other registered RIAs	33.1	53.8
Other Form 13F institutions	20.3	33.0
Households and other non-13F holders, residual	38.6	—
Panel D. Share of the U.S. green-ETF market by reporting sector, 2024Q4 ($91.7 billion)		
All registered RIAs	45.2	71.6
Household-serving single-county RIAs	12.8	20.3
Other registered RIAs	32.4	51.3
Other Form 13F institutions	17.9	28.4
Households and other non-13F holders, residual	36.9	—

Notes. This table reports the scale of the adviser ETF channel and its relevance for households. Advisers serving predominantly household accounts have a majority of individual clients; single-county advisers are represented in one U.S. county. Panel A covers registered investment advisers, including relying advisers, during 2009–2024. Panel B covers adviser-quarters with positive ETF holdings; aggregate ratios pool market values, while the median is across adviser-quarters. In Panels C and D, the first column uses December 2024 CRSP ETF market capitalization and the second uses CRSP-matched ETF value reported on Form 13F. Residuals equal market value not observed on Form 13F. Green ETFs follow the paper's classification.

TABLE C.2: ADVISER GREEN-ETF PORTFOLIO RESPONSE

	Dependent variable:					
	Green-ETF share response (bp)			Entry into green ETFs (pp)		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Advisers weighted by their share of the county ETF book						
Affected × Interpretation	23.21***	27.19***	26.81***	5.20**	5.06**	6.07**
	(7.76)	(8.25)	(8.94)	(2.23)	(2.53)	(2.41)
Affected × Attention	10.68**	11.02*	12.53*	−.42	.84	1.15
	(5.18)	(5.98)	(6.94)	(1.11)	(1.14)	(1.17)
Total Interpretation effect among affected advisers	22.81***	26.78***	26.57***	5.27**	5.13**	6.20**
	(7.73)	(8.24)	(9.00)	(2.20)	(2.50)	(2.40)
Observations	465,701	465,701	465,641	350,373	350,373	350,322
R^2	.090	.090	.132	.196	.196	.375
Affected adviser-episodes	1,156	1,156	1,096	857	857	806
Panel B. Advisers weighted equally						
Affected × Interpretation	14.55**	14.12**	15.06**	2.52*	3.32**	4.00**
	(6.97)	(6.83)	(7.52)	(1.30)	(1.43)	(1.61)
Affected × Attention	15.30***	16.71***	20.55***	−.58	.27	.64
	(3.76)	(4.02)	(4.83)	(.95)	(1.07)	(1.24)
Total Interpretation effect among affected advisers	14.37**	13.95**	14.65*	2.45*	3.24**	3.94**
	(6.90)	(6.77)	(7.46)	(1.31)	(1.44)	(1.61)
Observations	465,701	465,701	465,641	350,373	350,373	350,322
R^2	.018	.018	.155	.075	.075	.346
Affected adviser-episodes	1,156	1,156	1,096	857	857	806
Episode and quarter FE	✓	✓	✓	✓	✓	✓
County FE	✓	✓		✓	✓	
Adviser FE			✓			✓
Affected × hazard × year FE	✓	✓	✓	✓	✓	✓
Financial and demographic controls	✓	✓	✓	✓	✓	✓
Own-county damage, deaths and injuries	✓	✓	✓	✓	✓	✓
Political and belief controls	✓	✓	✓	✓	✓	✓
Lagged climate attention	✓	✓	✓	✓	✓	✓
Affected × controls		✓	✓		✓	✓

Notes. See notes to Table 2. The unit is the adviser-episode. In columns 1–3 the outcome is the change in the adviser green-ETF share from the quarter before the episode to the quarter after it, at pre-event prices, in basis points. In columns 4–6 the sample is restricted to adviser-episodes with no green ETF in the quarter before the episode, and the outcome is 100 times an indicator for a positive green-ETF weight in the quarter after it, at pre-event prices (a linear probability model; a positive weight is a weight above 10^{-8}). Panel A weights each adviser-episode by its share of the county ETF book in the quarter before the episode; Panel B weights advisers equally. Financial controls are the logs of the adviser ETF book and of its total Form 13F holdings in the quarter before the episode; Affected, Interpretation, Attention and the remaining controls are those of the county. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

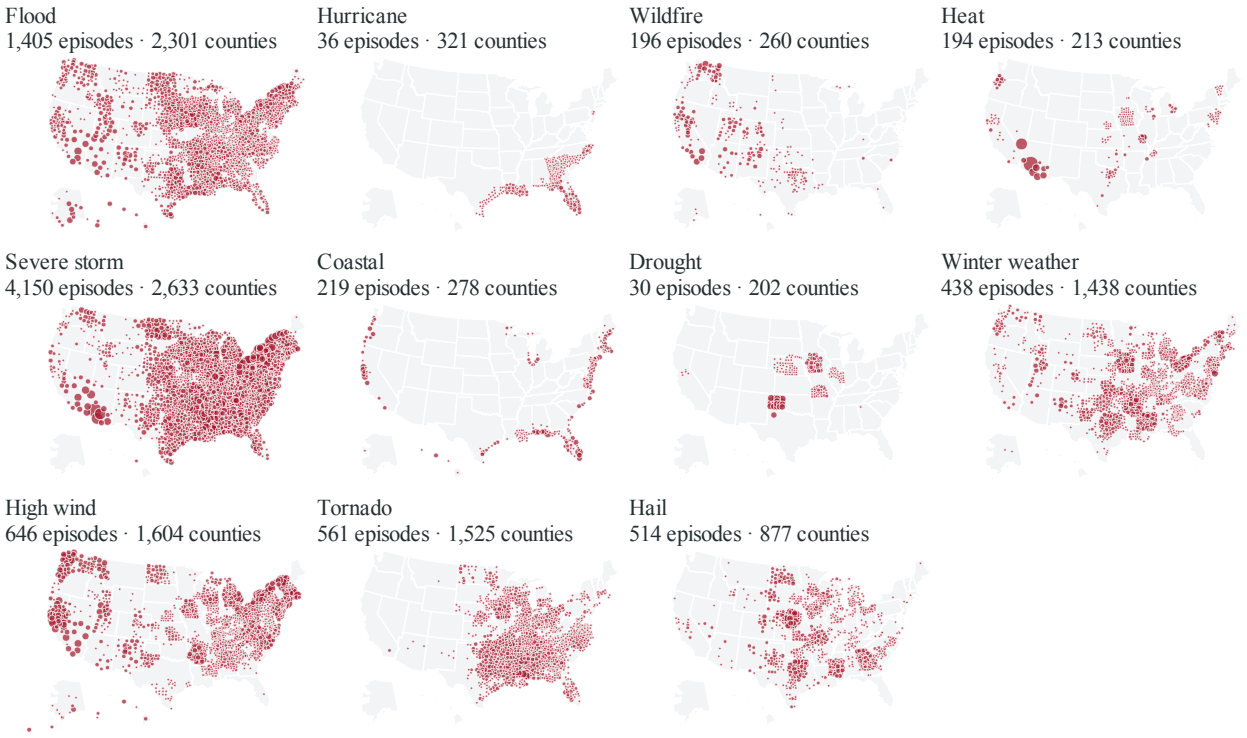


FIGURE C.1: GEOGRAPHIC DISTRIBUTION OF WITHIN-HAZARD TOP-SALIENCE EPISODES

Notes: The maps show U.S. counties affected by climate-hazard episodes beginning in 2009–2024. Point area is the county-level number of eligible episodes. The maps keep episodes ranking in the top 15 percent of salience within their hazard type.

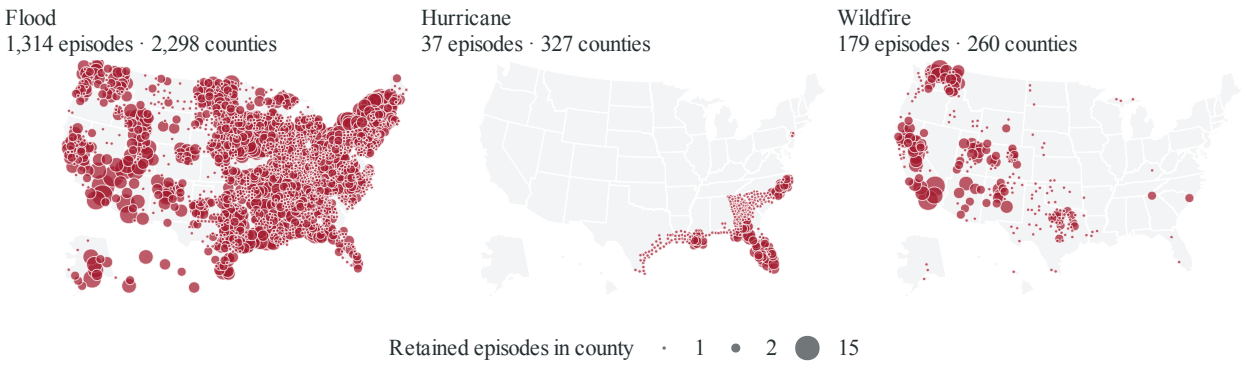


FIGURE C.2: GEOGRAPHIC DISTRIBUTION OF SELECTED FLOOD, HURRICANE, AND WILDFIRE EPISODES

Notes: The maps show the 2009–2024 county distribution of flood, hurricane, and wildfire episodes retained in the non-overlapping analysis samples. Point area is the retained episode count. Floods are taken from the joint three-hazard construction so that all panels share one sample definition.

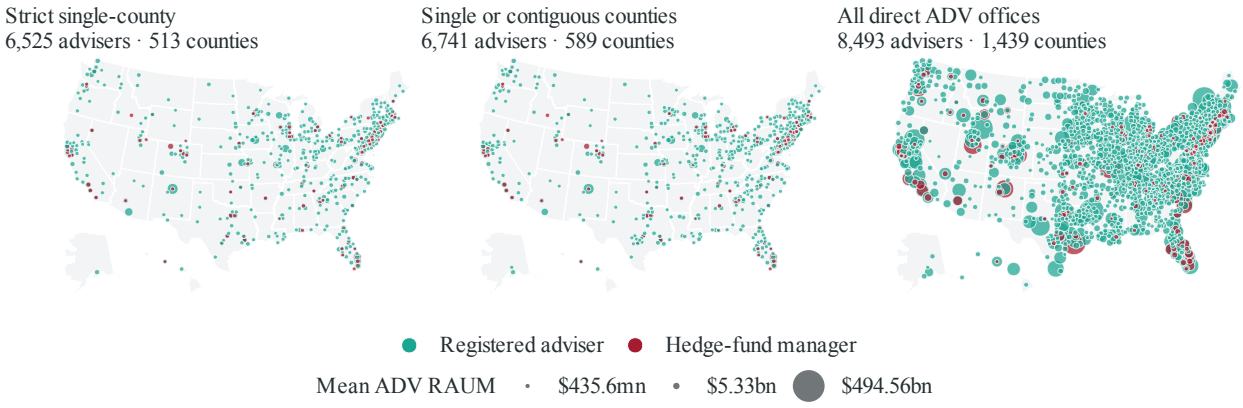


FIGURE C.3: LOCAL INVESTMENT-ADVISER AND HEDGE-FUND GEOGRAPHY

Notes: The maps show direct Form ADV principal or branch offices observed in 2009–2024. Teal denotes full registered investment advisers that are not hedge-fund managers; non-hedge-fund exempt reporting advisers are excluded. Red denotes hedge-fund managers with either a full registered-adviser or exempt-reporting-adviser Form ADV filing. The panels expand from a strict single-county footprint, to a footprint contained in connected counties, to all direct offices. Point area is mean Form ADV regulatory assets under management within the displayed county, panel, and investor-type cell. Office location is not client residence.

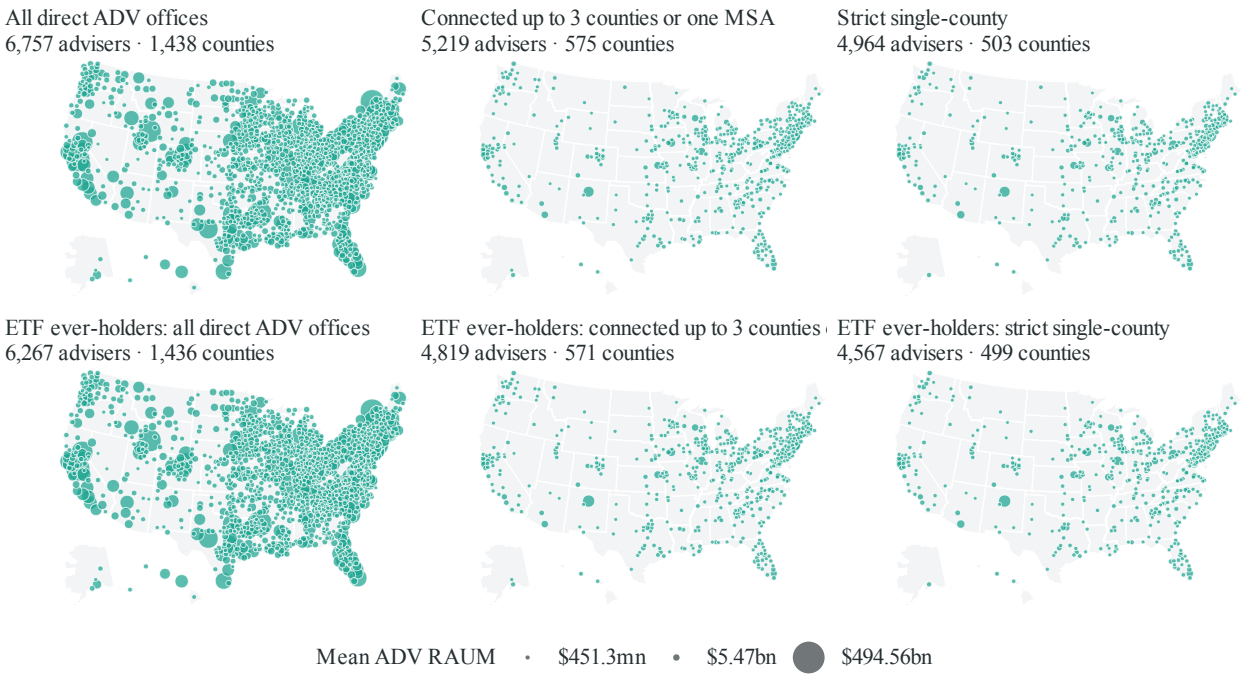


FIGURE C.4: REGISTERED INVESTMENT-ADVISER GEOGRAPHY AND ETF PARTICIPATION

Notes: See notes to Figure C.3. The maps here include full registered investment advisers only, excluding hedge-fund managers and exempt reporting advisers. From left to right, the top row shows all direct Form ADV offices, footprints contained in at most three connected counties or one metropolitan statistical area, and strict single-county footprints. The bottom row repeats those panels for advisers that held a positive ETF position at least once during 2009–2024. The ETF screen is fixed over the descriptive window.

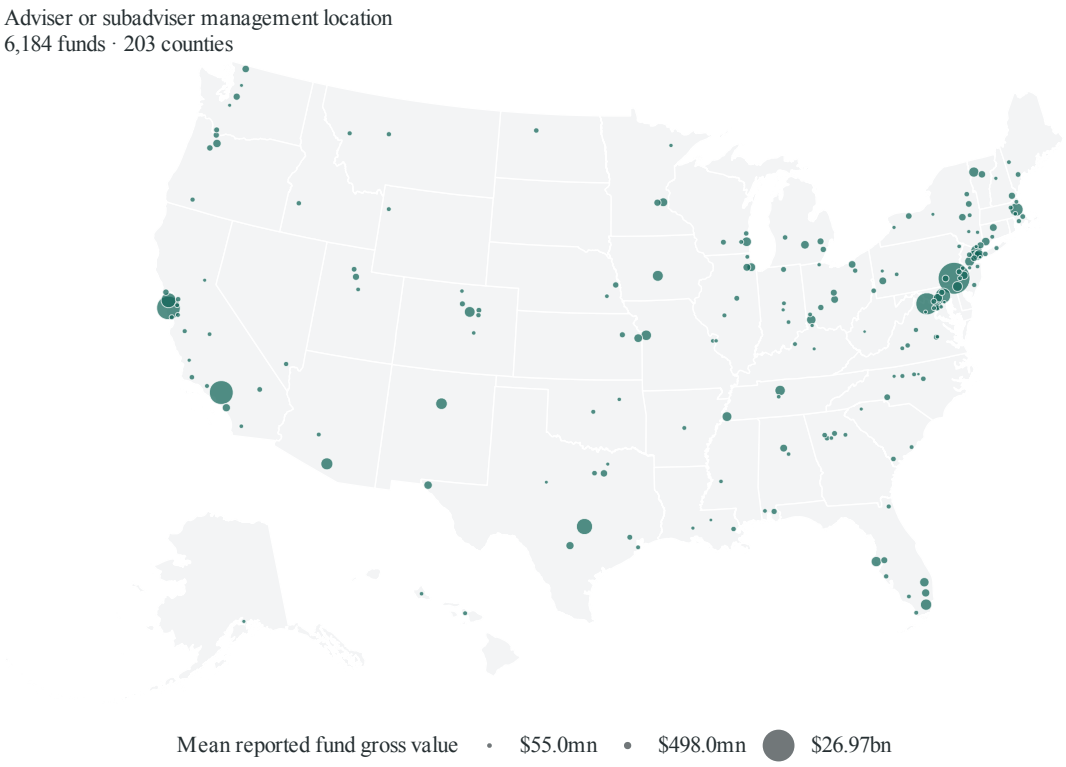


FIGURE C.5: MUTUAL-FUND MANAGEMENT GEOGRAPHY

Notes: The map places Thomson S12 funds at strict dated adviser or subadviser management counties observed in 2009–2024. Point area is mean reported fund gross value. Management location is not fund-share purchaser residence.

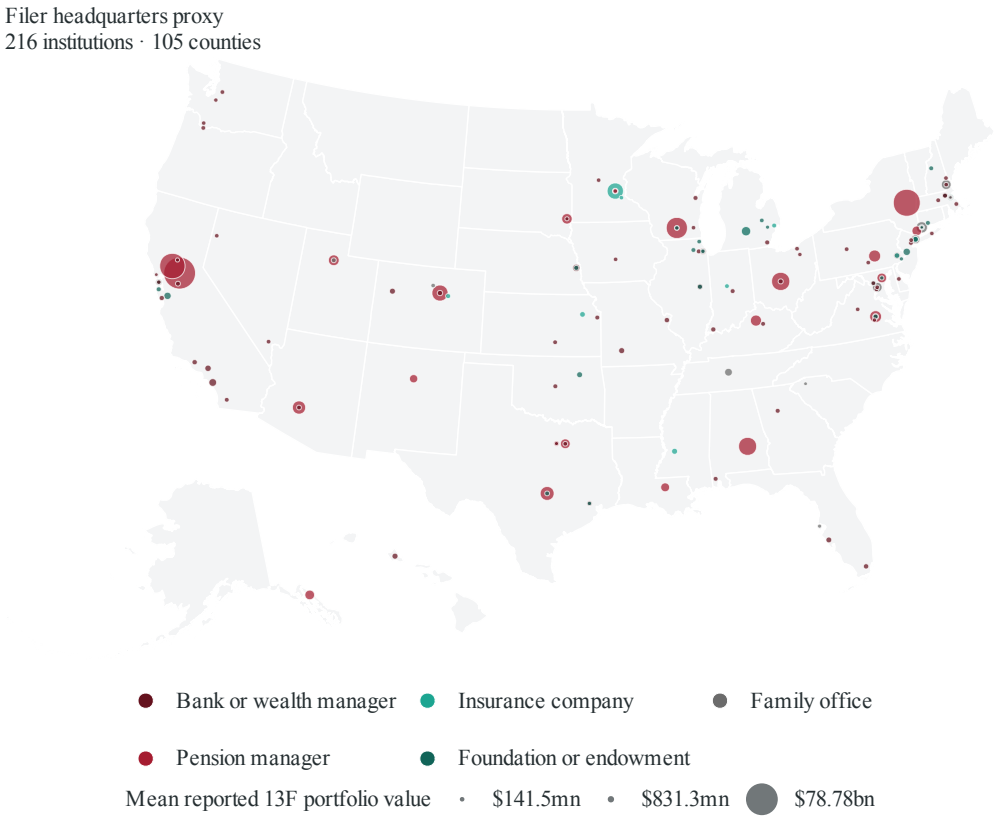
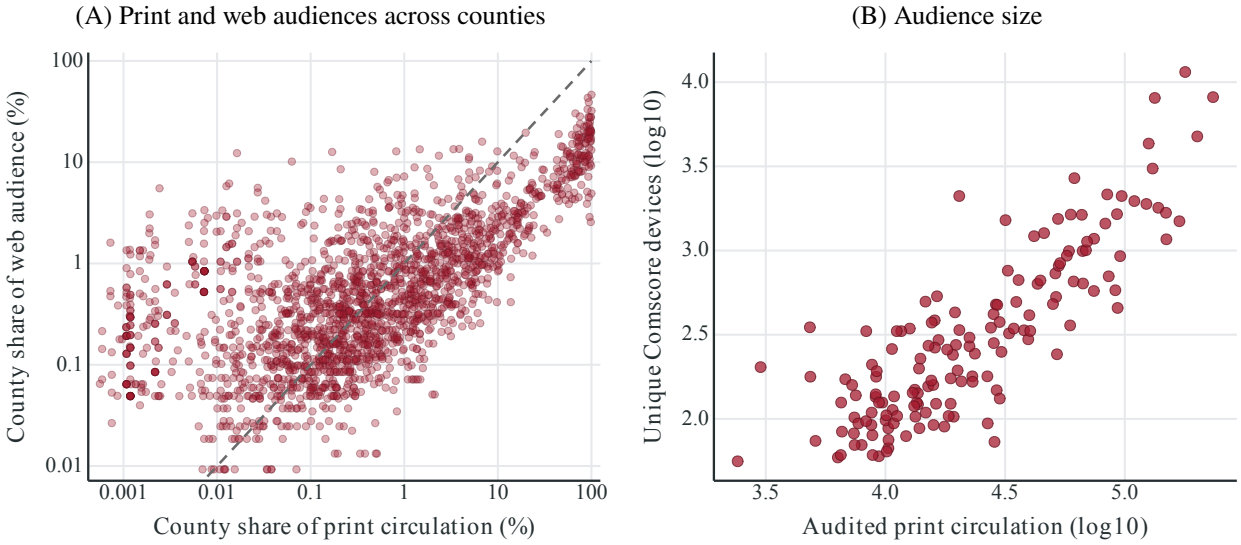


FIGURE C.6: INSTITUTIONAL HEADQUARTERS PROXIES

Notes: The map shows filer-level headquarters proxies for banks or wealth managers, pension managers, insurance companies, foundations or endowments, and family offices observed in 2009–2024. Color identifies institution type and point area is mean reported 13F portfolio value within the displayed county-type cell. Headquarters is a descriptive institutional location, not a verified portfolio-manager or beneficial-owner residence.

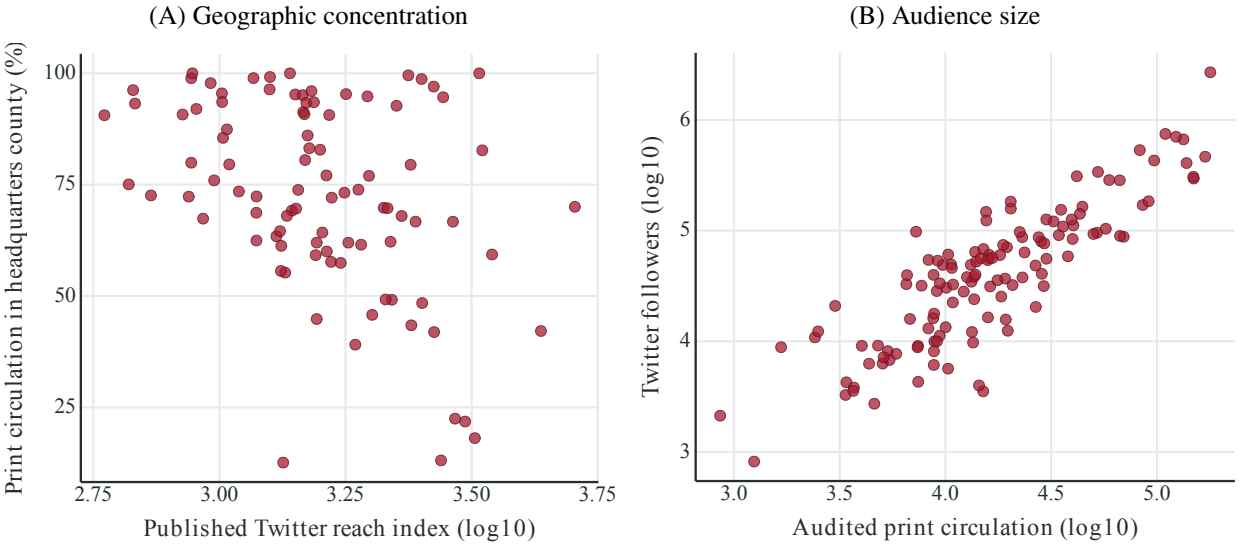
D Appendix to Section 2.2

FIGURE D.1: PRINT CIRCULATION AND OBSERVED WEB AUDIENCES



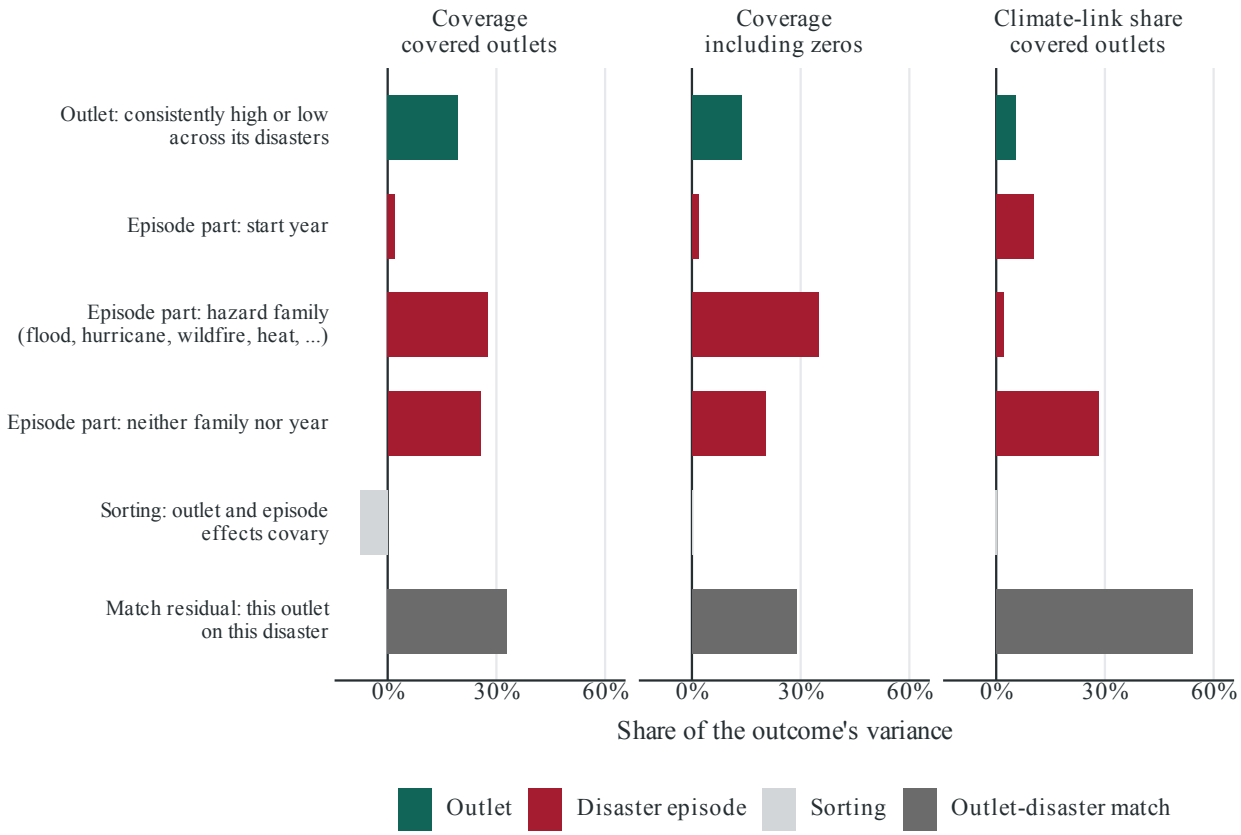
Notes. This figure plots observed Comscore web audiences against independently audited print circulation for 153 local newspapers. Panel A displays 2,114 newspaper–county pairs with positive observations in both sources on logarithmic axes labelled in percentages. Each share divides the county's audience by that newspaper's audience across all observed counties in the respective source; shares are computed before restricting to common counties. All observations are retained, with no geographic trim applied to these raw web shares. The dashed line denotes equal shares. The Pearson correlation of the untransformed shares is 0.822 and the Spearman correlation is 0.628. Panel B compares audited circulation with distinct outlet-wide Comscore devices (Spearman correlation 0.823). Comscore covers 2020; print averages available audits during 2015–2025. Newspapers require at least 50 unique web devices. Ambiguous shared print-audit mappings and one newspaper with invalid negative audited circulation are excluded. Web observations count distinct panel devices within each newspaper–county pair over the year, regardless of visit frequency; a device can appear in multiple counties. Common counties account for an average 96.1% of observed print circulation and 25.5% of observed county web-device counts per newspaper.

FIGURE D.2: PRINT CIRCULATION AND TWITTER AUDIENCES



Notes. This figure plots published Twitter audience measures against audited print circulation for matched local publications. Panel A compares the published population-reach index of Le Quéré, Chiang and Naaman (2022) with the share of observed audited circulation in the newspaper's headquarters county (93 newspapers; Spearman correlation −0.360). Higher index values indicate a geographically broader Twitter audience. Panel B compares total Twitter followers with total observed audited print circulation (127 newspapers; Spearman correlation 0.839). Each newspaper receives equal weight. Twitter counts are recorded in 2020 or 2022; the print footprint averages available audits during 2015–2025. These snapshots are used as fixed audience proxies. The county-share denominator is the newspaper's observed audited circulation across all available counties. Ambiguous publication matches, conflicting headquarters locations in Panel A, and one publication with invalid negative audited circulation are excluded.

FIGURE D.3: VARIANCE DECOMPOSITION OF COVERAGE AND CLIMATE FRAMING



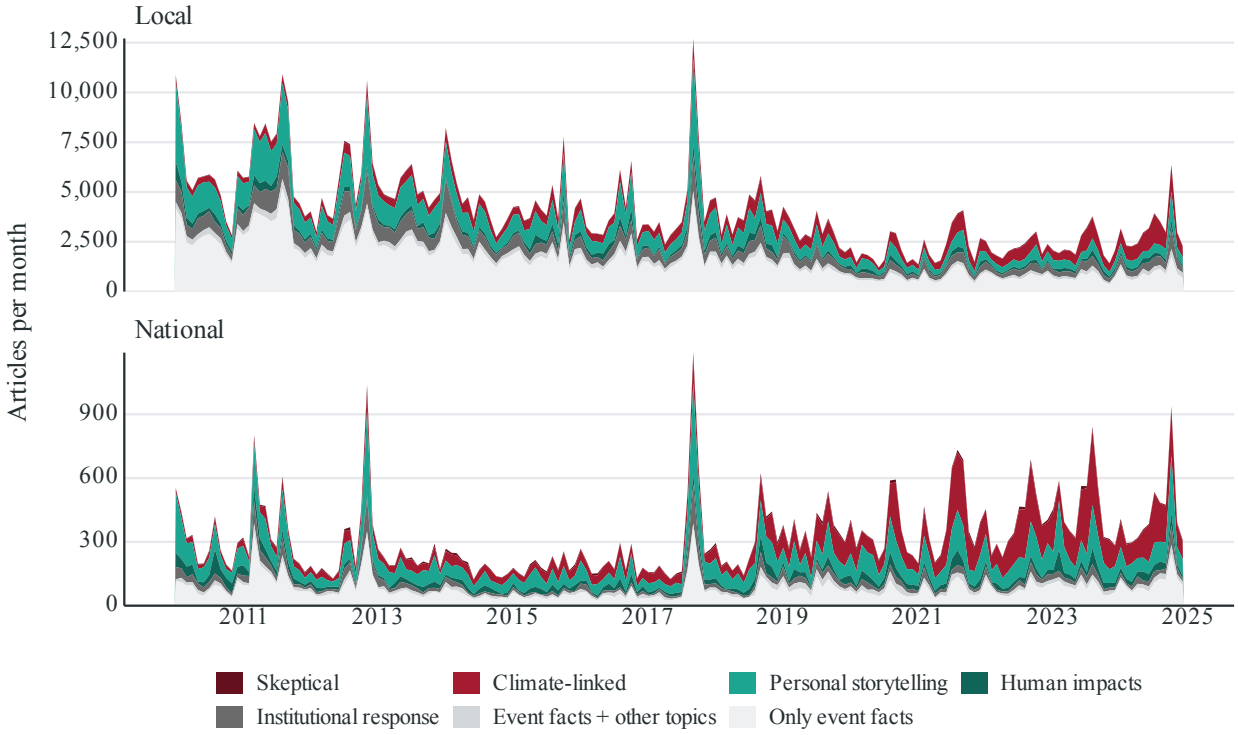
Notes. This figure reports the variance decomposition of coverage and of the climate framing share over outlet by event cells (hazard-matched articles, weeks 0 to 3 from the event start) into an outlet component, an event component, their sorting covariance, and the outlet-event match. Coverage is the log article count among covered outlets; framing is the climate-link share among covered outlets. Components are shares of total variance.

TABLE D.1: COMPOSITION OF DISASTER COVERAGE: CHAIN-OWNED AND INDEPENDENT LOCAL OUTLETS

	Dependent variable: article belongs to the category (percent)						
	Skeptical	Climate-linked	Personal storytelling	Human impacts	Institutional response	Event facts + other topics	Only event facts
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. No fixed effects							
Local, chain-owned	−0.66***	−4.22**	0.31	1.01*	0.24	−0.85	4.16*
	(0.24)	(2.10)	(2.00)	(0.57)	(0.91)	(1.52)	(2.15)
Constant	1.54	21.15	20.64	4.90	12.86	6.55	32.36
	(0.23)	(2.07)	(1.98)	(0.54)	(0.86)	(1.51)	(1.97)
Observations	383,872	383,872	383,872	383,872	383,872	383,872	383,872
R^2	.0003	.0009	.0000	.0001	.0000	.0001	.0005
Panel B. Calendar-quarter fixed effects							
Local, chain-owned	−0.67***	−3.83**	0.30	1.02*	0.26	−0.90	3.81*
	(0.23)	(1.66)	(1.76)	(0.58)	(0.90)	(1.51)	(2.14)
Observations	383,872	383,872	383,872	383,872	383,872	383,872	383,872
R^2	.0016	.0333	.0103	.0046	.0021	.0019	.0093
Calendar-quarter FEs	✓	✓	✓	✓	✓	✓	✓

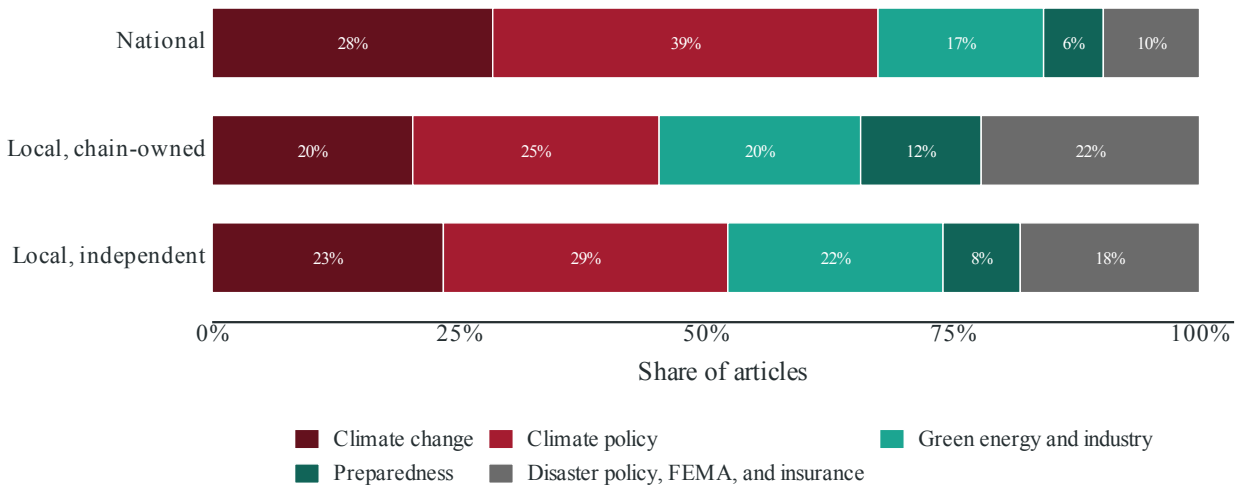
Notes. Each column is one regression. The unit is a disaster article published by a local outlet from 2015 through 2024. The dependent variable equals 100 when the article belongs to the category named in the column header and zero otherwise; the categories are those of the composition figure and are mutually exclusive. Local, chain-owned equals one when the outlet's owner in the quarter of publication holds two or more newspaper titles, and zero when it holds one, so the constant in Panel A is the category's share among independent outlets. Panel B adds calendar-quarter fixed effects. The sample has 370 outlets. Standard errors, in parentheses, are clustered by outlet. Stars are reported for the chain coefficient: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

FIGURE D.4: MONTHLY COMPOSITION OF DISASTER COVERAGE



Notes. See notes to Figure 2. Stacks report monthly publication-instance counts from 2010 through 2024, and the local and national panels use separate vertical scales.

FIGURE D.5: CLIMATE AND DISASTER-RELATED COVERAGE BEYOND INDIVIDUAL EVENTS BY OUTLET OWNERSHIP



Notes. This figure plots the composition of relevant articles from 2015 through 2024 that are not centred on a disaster event, for national outlets, chain-owned local outlets and independent local outlets; such articles may still mention disasters. Shares are calculated among 413,255 articles: 47,911 national, 324,050 chain-owned local and 41,294 independent local. Outlet classes are defined in the notes to Figure 2. Categories are mutually exclusive, and disaster policy, FEMA and insurance are combined as institutional management of disaster risk.

E Appendix to Section 3

TABLE E.1: LOCAL CLIMATE FRAMING AND GREEN ETF DEMAND AFTER FLOODING DISASTERS, CALENDAR-QUARTER TIMING

	(1)	(2)	(3)	(4)	(5)
Panel A. Fixed-price green ETF share change (basis points)					
Direct hit × framing (1 SD)	42.07**	43.89**	46.76**	46.76**	47.92**
	(18.14)	(18.76)	(20.51)	(20.50)	(20.67)
Climate framing (1 SD)	−6.71	−6.87	−7.47	−7.46	−7.20
	(4.23)	(4.25)	(4.38)	(4.39)	(4.34)
Direct hit	13.63	15.61	9.04	9.04	12.86
	(27.20)	(26.65)	(27.05)	(27.06)	(28.18)
R^2	.094	.103	.105	.105	.106
Mean dependent variable (bp)	1.06	1.06	1.06	1.06	1.06
Panel B. Current-price green ETF share change (basis points)					
Direct hit × framing (1 SD)	42.62**	44.47**	47.43**	47.41**	48.92**
	(18.19)	(18.78)	(20.58)	(20.56)	(20.84)
Climate framing (1 SD)	−6.74	−6.89	−7.51	−7.45	−7.19
	(4.20)	(4.21)	(4.34)	(4.35)	(4.30)
Direct hit	13.51	15.53	8.85	8.86	12.55
	(27.84)	(27.29)	(27.60)	(27.54)	(28.55)
R^2	.098	.107	.109	.109	.111
Mean dependent variable (bp)	1.21	1.21	1.21	1.21	1.21
Share observations	77,571	77,571	77,571	77,571	77,571
Counties	319	319	319	319	319
Disaster episodes	632	632	632	632	632
County fixed effects	✓	✓	✓	✓	✓
Calendar-quarter fixed effects	✓	✓	✓	✓	✓
Disaster-episode fixed effects	✓	✓	✓	✓	✓
Year-specific direct-hit effects	✓	✓	✓	✓	✓
Prior green-share change		✓	✓	✓	✓
Lagged portfolio assets			✓	✓	✓
Predetermined political controls				✓	✓
Predetermined demographic controls					✓

Notes. See notes to Table E.2. Climate framing is measured over the event's calendar quarter. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table E.2: County green-ETF portfolio response

	(1)	(2)	(3)	(4)	(5)
Panel A. Fixed prices					
Hit × climate framing	36.01*** (13.20)	38.71*** (13.06)	38.73*** (13.05)	38.80*** (12.99)	43.05*** (13.26)
Affected	-8.88 (48.92)	-6.11 (47.97)	-13.74 (48.64)	-14.04 (48.10)	-16.39 (50.19)
Climate framing	-4.50* (2.44)	-4.62* (2.50)	-4.66* (2.47)	-4.59* (2.47)	-4.08* (2.44)
News attention	-12.72** (5.48)	-13.26** (5.58)	-12.66** (5.57)	-12.68** (5.58)	-12.78** (5.85)
Hit × news attention	33.21* (17.64)	34.13* (18.58)	33.94* (19.16)	34.16* (19.11)	34.82* (20.09)
Observations	59,330	59,330	59,330	59,330	59,330
Panel B. Observed prices					
Hit × climate framing	34.78*** (13.12)	37.46*** (12.97)	37.48*** (12.96)	37.62*** (12.85)	42.56*** (13.39)
Affected	-10.64 (51.07)	-7.89 (50.11)	-15.61 (50.71)	-16.16 (50.04)	-17.83 (51.66)
Climate framing	-4.81** (2.43)	-4.93** (2.48)	-4.97** (2.46)	-4.85* (2.46)	-4.34* (2.41)
News attention	-13.39** (5.55)	-13.92** (5.65)	-13.31** (5.64)	-13.35** (5.65)	-13.52** (5.91)
Hit × news attention	34.30* (18.55)	35.22* (19.46)	35.02* (20.05)	35.43* (20.01)	35.76* (20.93)
Observations	59,330	59,330	59,330	59,330	59,330
County fixed effects	Yes	Yes	Yes	Yes	Yes
Quarter and episode fixed effects	Yes	Yes	Yes	Yes	Yes
Year-specific direct-hit effects	Yes	Yes	Yes	Yes	Yes
Prior green-share change		Yes	Yes	Yes	Yes
Lagged log portfolio assets			Yes	Yes	Yes
Political controls				Yes	Yes
Demographic controls					Yes

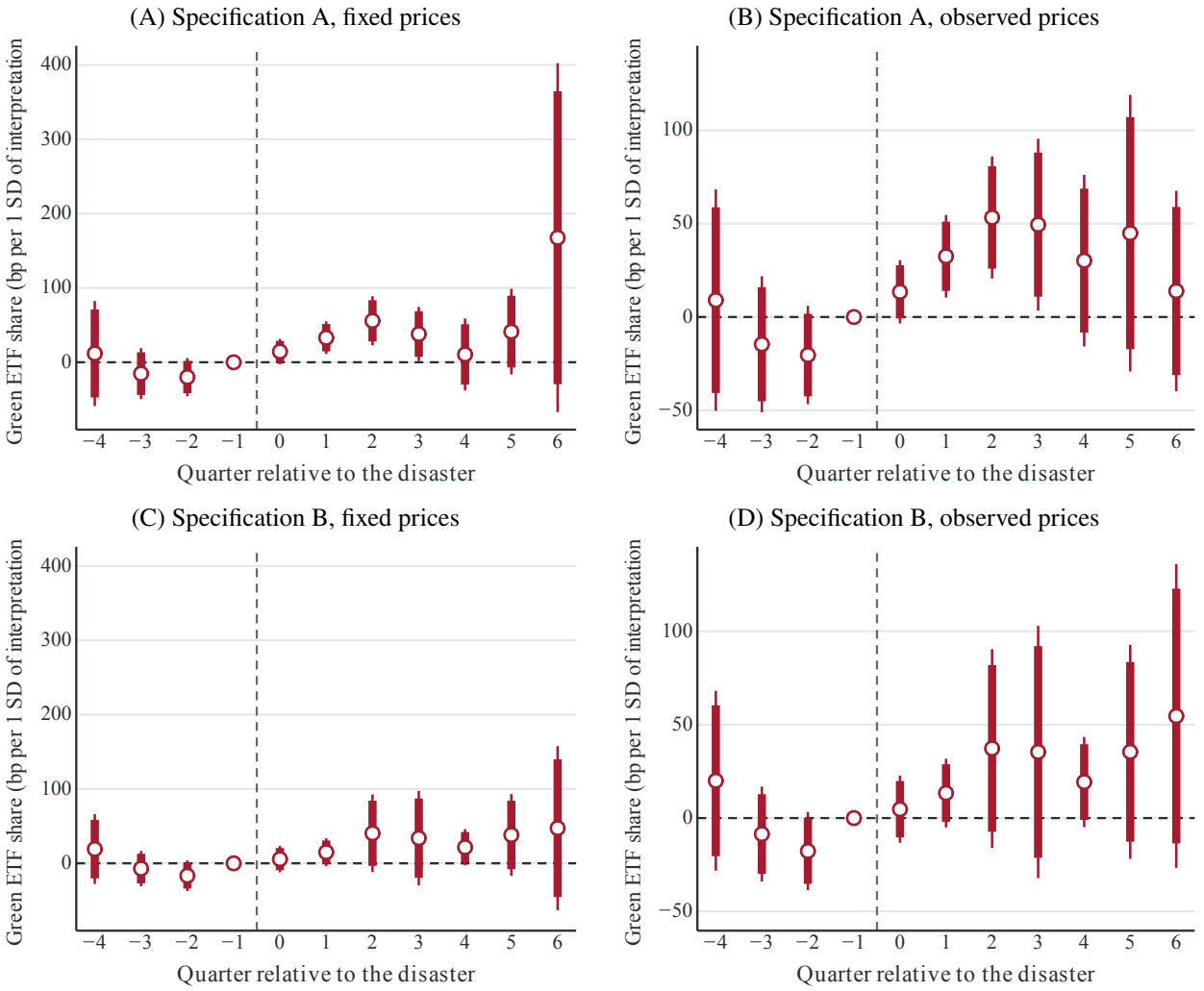
Notes: The outcome is the change in green-ETF share from $t-1$ to $t+1$, in basis points. Framing and log total disaster-news attention both use event weeks 0–3 and are standardized. The sample covers high-salience flood, hurricane and coastal episodes, 2018Q1–2024Q4. Standard errors are clustered by county and episode, retaining the original covariance convention. Standard errors are in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE E.3: ADVISER-LEVEL LOCAL CLIMATE FRAMING AND GREEN ETF DEMAND AFTER FLOODING DISASTERS, CALENDAR-QUARTER TIMING

	(1)	(2)	(3)	(4)
Panel A. Fixed-price green ETF share change (basis points)				
Direct hit × framing (1 SD)	10.41*	10.11*	10.05*	10.07*
	(5.39)	(5.68)	(5.69)	(5.67)
Climate framing (1 SD)	−1.05	−1.19	−1.17	−1.32
	(1.52)	(1.53)	(1.54)	(1.55)
Direct hit	1.95	2.27	2.23	.77
	(4.24)	(4.71)	(4.69)	(4.88)
R^2	.069	.144	.144	.144
Mean dependent variable (bp)	−.62	−.62	−.62	−.62
Panel B. Current-price green ETF share change (basis points)				
Direct hit × framing (1 SD)	10.48**	10.19*	10.13*	10.15*
	(5.29)	(5.55)	(5.56)	(5.55)
Climate framing (1 SD)	−.84	−.97	−.95	−1.11
	(1.73)	(1.74)	(1.75)	(1.74)
Direct hit	1.16	1.46	1.42	−.15
	(4.47)	(4.90)	(4.89)	(5.15)
R^2	.067	.133	.133	.133
Mean dependent variable (bp)	−.58	−.58	−.58	−.58
Share observations	621,761	621,761	621,761	621,761
Advisers	2,765	2,765	2,765	2,765
Disaster episodes	632	632	632	632
Adviser fixed effects	✓	✓	✓	✓
Calendar-quarter fixed effects	✓	✓	✓	✓
Disaster-episode fixed effects	✓	✓	✓	✓
Year-specific direct-hit effects	✓	✓	✓	✓
Prior green-share change		✓	✓	✓
Lagged portfolio assets			✓	✓
Predetermined political controls				✓

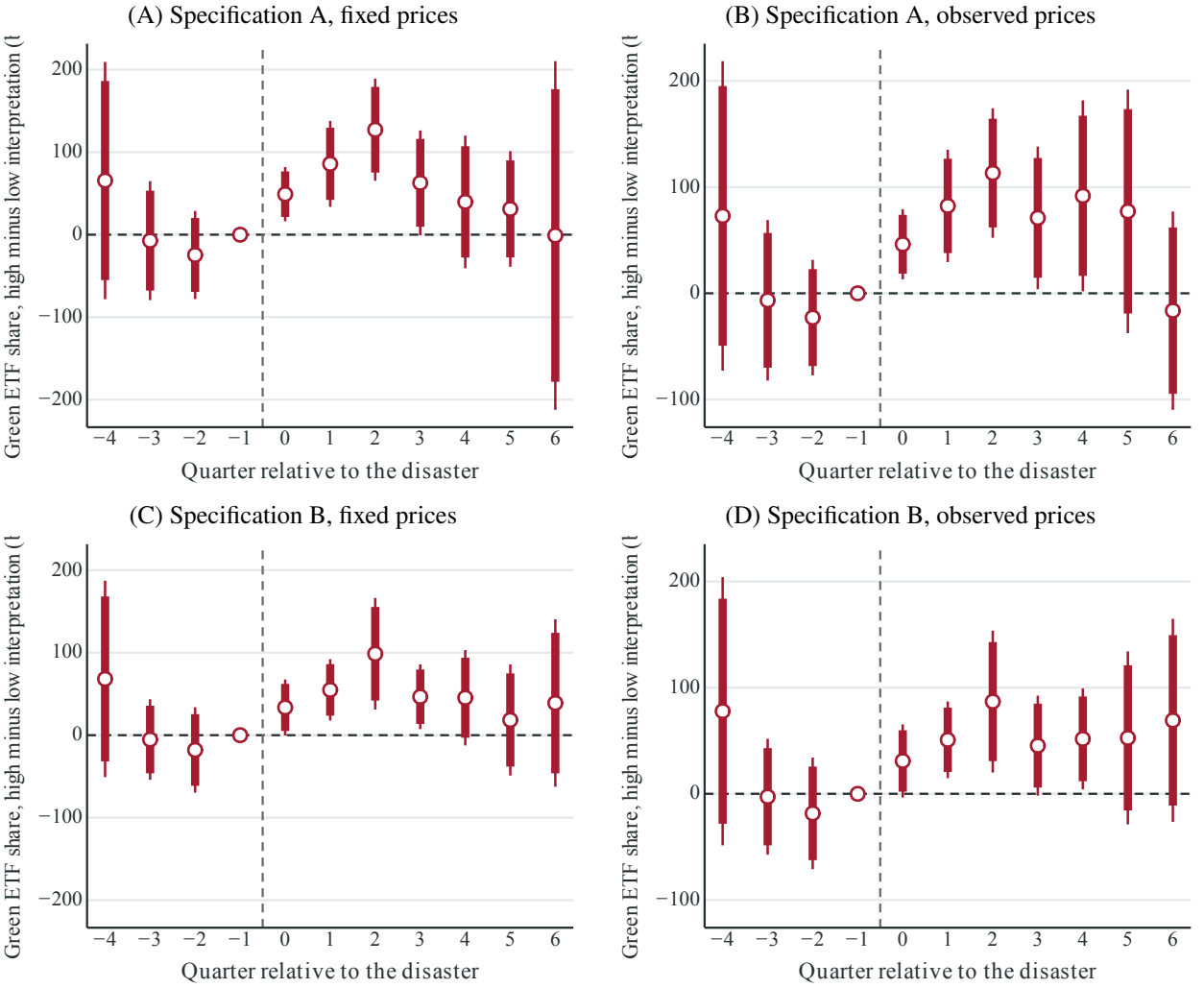
Notes. This table reports the adviser-level green-ETF share response to the climate framing of local disaster coverage, after flood, hurricane and coastal episodes. Framing is the circulation-weighted average across local outlets of the outlet-quarter share of disaster-related articles that connect the disaster to climate change. Climate framing is the slope among linked non-hit observations; Direct hit is the hit difference in the reference year at mean framing and coverage; their interaction is the additional framing slope among directly hit observations. Share outcomes equal $Y_{t+1} - Y_{t-1}$. Panel A holds prices fixed at pre-event values; Panel B uses observed prices. Column (1) includes year-specific direct-hit effects. Column (2) adds the immediately preceding green-share change; column (3) adds lagged log portfolio assets; column (4) adds predetermined political controls. Adviser-level demographics are not included. All columns include coverage, Direct hit by coverage, adviser, calendar-quarter, and disaster-episode fixed effects, and standard errors clustered by adviser location. R^2 includes absorbed fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

FIGURE E.1: FLOODS, HURRICANES AND COASTAL EPISODES: ONE-STANDARD-DEVIATION FRAMING RESPONSE. THE SAMPLE BEGINS IN 2015. POINT ESTIMATES AND POINTWISE 90% AND 95% CONFIDENCE INTERVALS ARE SHOWN; THE PLOTTED NOTES DEFINE EACH COMPARISON.



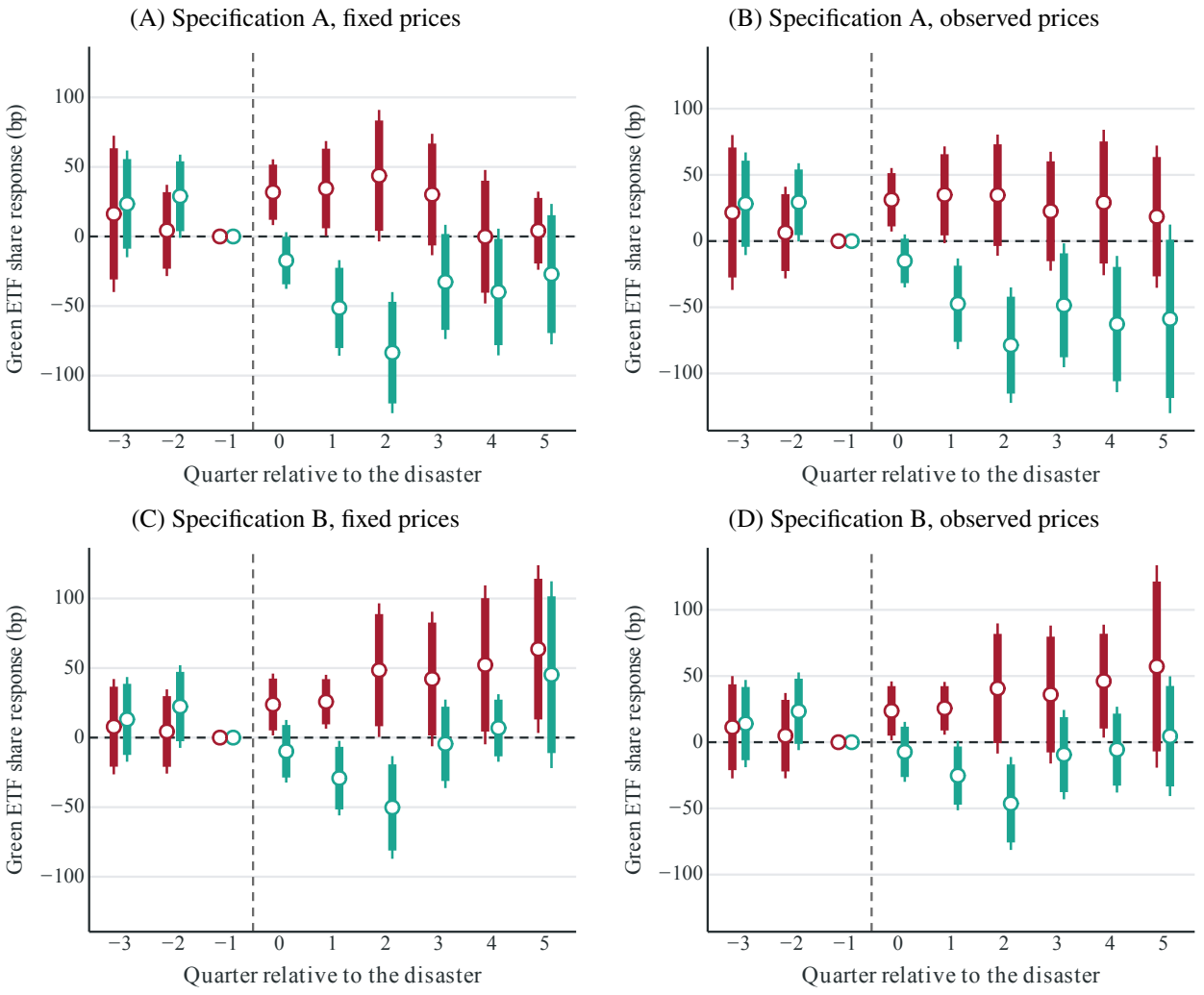
Notes. Thin outer bars: 95% CI. Thick inner bars: 90% CI. Pointwise normal intervals; two-way clustering by county and disaster episode. Specification A: county, calendar-quarter and disaster-episode FE. Specification B: calendar-quarter and disaster-episode FE, without county FE. Both specifications include Hit × Year terms. Prior green-share change is excluded. Controls include lagged assets, prior politics, demographics, framing and attention. Both prices span −4 to +6; quarter −1 is normalized to zero. Conditional Hit × Framing coefficient per one SD on the common frozen scale. Joint pretest: quarters −4, −3, −2; p = 0.230 (A), 0.305 (B), 0.208 (C), 0.151 (D).

FIGURE E.2: FLOODS, HURRICANES AND COASTAL EPISODES: DIFFERENCE BETWEEN HIGH AND LOW FRAMING DISASTER RESPONSES. THE SAMPLE BEGINS IN 2015. POINT ESTIMATES AND POINTWISE 90% AND 95% CONFIDENCE INTERVALS ARE SHOWN; THE PLOTTED NOTES DEFINE EACH COMPARISON.



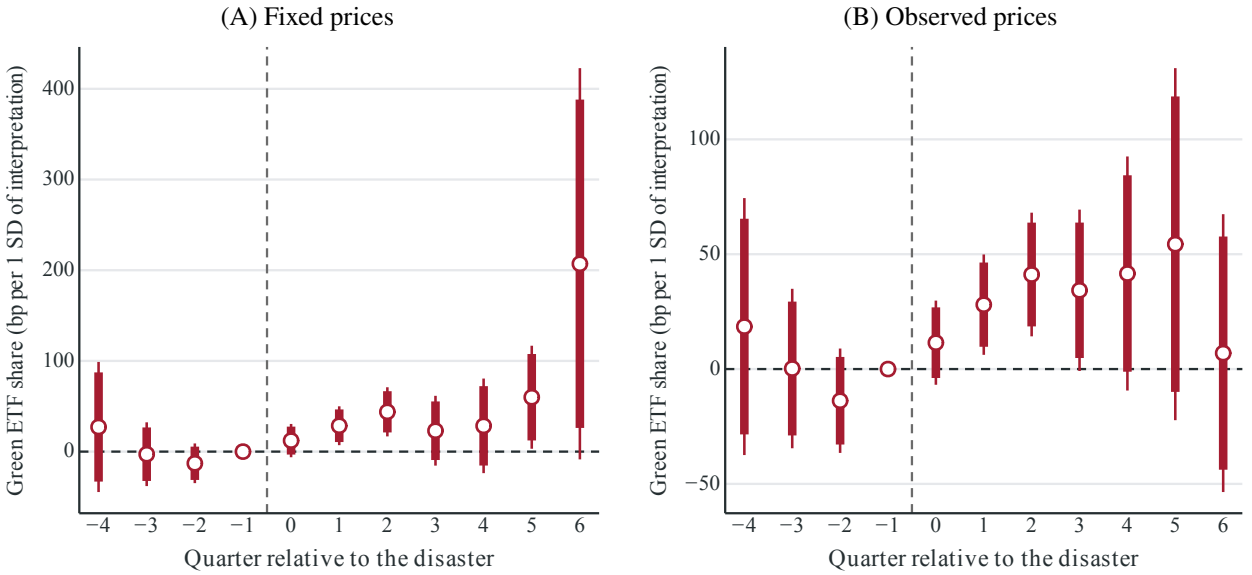
Notes. Thin outer bars: 95% CI. Thick inner bars: 90% CI. Pointwise normal intervals; two-way clustering by county and disaster episode. Specification A: county, calendar-quarter and disaster-episode FE. Specification B: calendar-quarter and disaster-episode FE, without county FE. Both specifications include Hit × Year terms. Prior green-share change is excluded. Controls include lagged assets, prior politics, demographics, framing and attention. Both prices span −4 to +6; quarter −1 is normalized to zero. Estimated difference in conditional disaster responses between High and Low framing. Joint pretest tests this gap at −4, −3, −2; $p = 0.366$ (A), 0.325 (B), 0.463 (C), 0.397 (D). Median cutoff is fixed across horizons and specifications. Group comparisons retain the continuous framing main effect.

FIGURE E.3: FLOODS, HURRICANES AND COASTAL EPISODES: CONDITIONAL DISASTER RESPONSES BY FRAMING GROUP. THE SAMPLE BEGINS IN 2015. POINT ESTIMATES AND POINTWISE 90% AND 95% CONFIDENCE INTERVALS ARE SHOWN; THE PLOTTED NOTES DEFINE EACH COMPARISON.



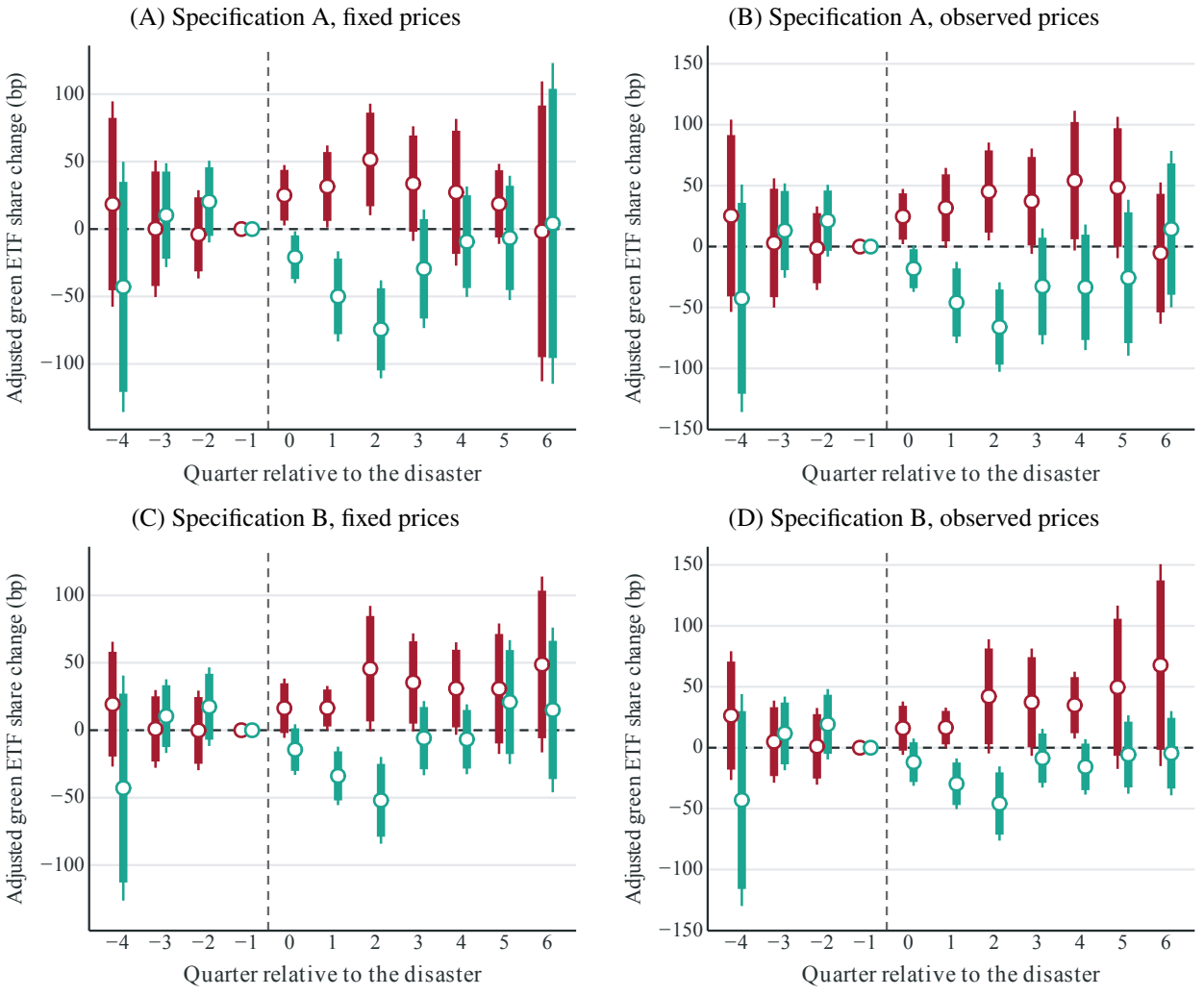
Notes. Thin outer bars: 95% CI. Thick inner bars: 90% CI. Pointwise normal intervals; two-way clustering by county and disaster episode. Specification A: county, calendar-quarter and disaster-episode FE. Specification B: calendar-quarter and disaster-episode FE, without county FE. Both specifications include Hit × Year terms. Prior green-share change is excluded. Controls include lagged assets, prior politics, demographics, framing and attention. Both prices span −3 to +5; quarter −1 is normalized to zero. Conditional disaster responses for each framing group, evaluated at a common reference. The joint pretest concerns the High–Low gap; $p = 0.366$ (A), 0.325 (B), 0.463 (C), 0.397 (D). Median cutoff is fixed across horizons and specifications. Group comparisons retain the continuous framing main effect.

FIGURE E.4: FLOODS, HURRICANES, COASTAL EPISODES AND WILDFIRES: ONE-STANDARD-DEVIATION FRAMING RESPONSE. THE SAMPLE BEGINS IN 2015. POINT ESTIMATES AND POINTWISE 90% AND 95% CONFIDENCE INTERVALS ARE SHOWN; THE PLOTTED NOTES DEFINE EACH COMPARISON.



Notes. Thin outer bars: 95% CI. Thick inner bars: 90% CI. Pointwise normal intervals; two-way clustering by county and disaster episode. County, calendar-quarter and disaster-episode FE; Hit × Year × Hazard terms. Prior green-share change is excluded. The framing scale and outcome definitions match those of the flood, hurricane and coastal specifications. Controls include lagged assets, prior politics, demographics, framing and attention. Both prices span −4 to +6; quarter −1 is normalized to zero. Conditional Hit × Framing coefficient per one SD on the common frozen scale. Joint pretest: quarters −4, −3, −2; $p = 0.148$ (A), 0.309 (B).

FIGURE E.5: FLOODS, HURRICANES AND COASTAL EPISODES: HIGH AND LOW FRAMING AMONG HIT. THE SAMPLE BEGINS IN 2015. POINT ESTIMATES AND POINTWISE 90% AND 95% CONFIDENCE INTERVALS ARE SHOWN; THE PLOTTED NOTES DEFINE EACH COMPARISON.



Notes. Thin outer bars: 95% CI. Thick inner bars: 90% CI. Pointwise normal intervals; two-way county and episode clustering. Specification A: county, calendar-quarter and episode FE. Specification B: calendar-quarter and episode FE, without county FE. Both retain Hit × Year, framing, attention, Hit × Attention and remaining controls. Prior change is excluded. Both prices span −4 to +6; changes are relative to quarter −1, imposed at zero. Floods includes flood, hurricane and coastal events. Support varies by horizon. High/Low use the frozen median of M/A: the share of disaster reporting linking disasters to climate change. Framing is set to each group's weighted mean; attention, other controls and fixed effects share the within-horizon all-Hit reference. Both views use the same pooled median model. The joint pretest tests the between-series gap at −4, −3, −2, not either path separately; p = 0.368 (A), 0.327 (B), 0.468 (C), 0.439 (D).

TABLE E.4: COUNTY GREEN-HOLDING MARGINS

	(1)	(2)	(3)	(4)	(5)
Share of advisers holding green, all advisers (percentage points)					
Hit × climate framing	5.76*** (1.40)	5.77*** (1.40)	5.76*** (1.41)	5.74*** (1.44)	5.96*** (1.53)
Affected	3.27 (3.25)	3.27 (3.23)	2.87 (3.18)	2.99 (3.24)	2.95 (3.02)
Climate framing	-0.51* (0.29)	-0.51* (0.29)	-0.51* (0.29)	-0.53* (0.29)	-0.56* (0.29)
News attention	-0.54 (0.58)	-0.54 (0.58)	-0.51 (0.59)	-0.50 (0.59)	-0.58 (0.59)
Hit × news attention	-0.51 (1.28)	-0.50 (1.29)	-0.51 (1.31)	-0.60 (1.31)	-0.81 (1.23)
Observations	59,326	59,326	59,326	59,326	59,326
Share holding green, continuing in the same county (percentage points)					
Hit × climate framing	5.94*** (1.75)	5.94*** (1.74)	5.94*** (1.74)	5.91*** (1.75)	5.89*** (1.86)
Affected	2.75 (3.29)	2.75 (3.26)	2.50 (3.21)	2.51 (3.29)	2.22 (3.06)
Climate framing	-0.37 (0.28)	-0.37 (0.28)	-0.37 (0.28)	-0.40 (0.28)	-0.42 (0.28)
News attention	0.16 (0.52)	0.16 (0.52)	0.18 (0.52)	0.19 (0.52)	0.10 (0.53)
Hit × news attention	0.11 (1.14)	0.11 (1.15)	0.10 (1.17)	0.03 (1.16)	-0.14 (1.09)
Observations	56,693	56,693	56,693	56,693	56,693
County entry (percentage points)					
Hit × climate framing	1.64 (2.15)	1.55 (2.27)	1.46 (2.28)	1.42 (2.28)	0.99 (1.95)
Affected	4.22 (9.66)	4.38 (9.67)	4.83 (10.02)	5.52 (10.37)	3.66 (9.68)
Climate framing	0.53 (0.48)	0.54 (0.49)	0.54 (0.49)	0.51 (0.48)	0.52 (0.49)
News attention	0.54 (1.40)	0.54 (1.40)	0.52 (1.39)	0.51 (1.38)	0.60 (1.38)
Hit × news attention	-1.24 (2.76)	-1.40 (2.96)	-1.46 (2.95)	-1.71 (2.93)	-2.02 (2.80)
Observations	34,389	34,389	34,389	34,389	34,389
County incumbent share (basis points)					
Hit × climate framing	41.51** (16.00)	42.35*** (15.80)	32.62** (13.48)	32.26** (13.66)	44.82** (17.77)
Affected	-66.71 (59.78)	-57.57 (58.65)	-75.47** (37.51)	-86.67** (41.88)	-102.07* (56.50)
Climate framing	-10.26** (4.73)	-10.27** (4.97)	-11.06** (5.00)	-11.41** (5.03)	-10.52** (5.08)
News attention	-26.04** (10.28)	-27.23** (10.86)	-26.75** (11.39)	-26.32** (11.44)	-22.14* (12.43)
Hit × news attention	55.87*** (17.57)	53.63*** (18.19)	45.54** (18.89)	46.44** (19.12)	43.79** (21.09)
Observations	24,934	24,934	24,934	24,934	24,934
County fixed effects	Yes	Yes	Yes	Yes	Yes
Quarter and episode fixed effects	Yes	Yes	Yes	Yes	Yes
Year-specific direct-hit effects	Yes	Yes	Yes	Yes	Yes
Prior green-share change		Yes	Yes	Yes	Yes
Lagged log portfolio assets			Yes	Yes	Yes
Political controls				Yes	Yes
Demographic controls					Yes

Notes: Fixed-price outcomes from $t-1$ to $t+1$. Framing and log total disaster-news attention both use event weeks 0–3. The sample covers high-salience flood, hurricane and coastal episodes, 2018Q1–2024Q4. Entry is conditional on no pre-event green holdings; incumbent changes are conditional on positive pre-event green holdings. Share-of-adviser outcomes use endpoint-present advisers. County and episode clustering. Standard errors are in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE E.5: ADVISER GREEN-HOLDING MARGINS

	(1)	(2)	(3)	(4)
Adviser entry (percentage points)				
Hit × climate framing	3.95** (1.97)	3.90** (1.98)	3.90** (1.98)	3.89** (1.97)
Affected	-2.79 (3.13)	-2.72 (3.14)	-2.71 (3.14)	-2.54 (3.15)
Climate framing	-0.22 (0.15)	-0.22 (0.15)	-0.22 (0.15)	-0.18 (0.15)
News attention	0.30 (0.46)	0.28 (0.46)	0.28 (0.46)	0.30 (0.46)
Hit × news attention	0.61 (1.40)	0.56 (1.41)	0.56 (1.42)	0.54 (1.41)
Observations	248,833	248,833	248,833	248,833
Adviser incumbent share (basis points)				
Hit × climate framing	33.93 (24.62)	34.19 (24.70)	35.11 (24.59)	35.38 (24.44)
Affected	-97.35*** (27.66)	-97.04*** (27.49)	-99.63*** (28.39)	-100.54*** (28.52)
Climate framing	-2.22 (2.21)	-2.23 (2.18)	-2.25 (2.15)	-2.35 (2.20)
News attention	-0.10 (4.70)	0.09 (4.48)	0.43 (4.39)	0.07 (4.51)
Hit × news attention	64.16*** (18.93)	64.01*** (18.87)	63.89*** (18.64)	63.85*** (18.62)
Observations	105,185	105,185	105,185	105,185
Adviser fixed effects	Yes	Yes	Yes	Yes
Quarter and episode fixed effects	Yes	Yes	Yes	Yes
Year-specific direct-hit effects	Yes	Yes	Yes	Yes
Prior green-share change		Yes	Yes	Yes
Lagged log portfolio assets			Yes	Yes
Political controls				Yes

Notes: Fixed-price outcomes from $t-1$ to $t+1$. Framing and log total disaster-news attention both use event weeks 0–3. The sample covers high-salience flood, hurricane and coastal episodes, 2018Q1–2024Q4. Entry is conditional on no pre-event green holdings; incumbent changes are conditional on positive pre-event green holdings. Share-of-adviser outcomes use endpoint-present advisers. Clustering by adviser location county. Standard errors are in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

F Appendix to Section 3.4

TABLE F.1: WHO RESPONDS TO CLIMATE FRAMING: HETEROGENEITY OF THE FRAMING EFFECT

	Interactions with:			Surviving		
Characteristic	Affected	Framing	Hit × Framing	Hit × Framing	Affected	N
Panel A. Beliefs and ambient coverage						
Yale worried share	-13.9 (23.4)	-1.8 (2.7)	22.4* (13.3)	36.7*** (10.1)	-14.2 (45.8)	59,330
Perceived scientific consensus (Yale)	-0.7 (23.5)	-0.0 (2.8)	9.9 (16.4)	42.8*** (12.8)	-16.6 (46.9)	59,330
Policy talk (contemp.)	34.5** (17.1)	-5.3* (2.8)	4.3 (14.5)	32.6*** (12.1)	0.5 (54.3)	59,330
Policy talk (lagged)	28.6* (16.9)	-1.1 (2.5)	10.8 (15.0)	43.3*** (15.6)	-4.5 (53.3)	59,330
News penetration p.c.	18.7 (14.3)	-3.1 (3.0)	-0.2 (18.6)	43.1*** (13.5)	-21.5 (48.5)	59,330
Panel B. Politics and socioeconomic characteristics						
County Democratic vote	-8.2 (27.1)	-3.6 (2.8)	0.9 (11.0)	45.2*** (11.5)	-13.9 (44.9)	59,330
Median income	-29.0 (25.0)	0.6 (2.3)	17.5 (14.1)	47.2*** (15.0)	-28.7 (48.6)	59,330
College share	-28.2 (19.1)	0.6 (2.2)	4.6 (10.4)	39.3*** (12.7)	-19.9 (51.3)	59,330
Panel C. Exposure and experience						
Episode damage	-24.9* (13.5)	-0.1 (1.4)	10.1 (11.9)	41.1*** (12.7)	-12.7 (49.3)	59,330
Own-county damage	-28.3 (17.6)	—	13.1 (12.6)	41.0*** (12.2)	-9.2 (50.7)	59,330
Pre-2018 ETF turnover (high vs low)	100.4* (56.1)	-5.4 (6.8)	31.8 (53.2)	49.1*** (18.3)	45.8 (53.6)	43,558
Same-type history (0–60m)	4.3 (16.3)	-7.1 (5.2)	2.9 (11.0)	42.2** (18.6)	-27.4 (47.8)	59,330
No same-type history (0–60m)	-7.5 (56.6)	-6.8 (8.1)	124.8** (63.3)	42.2** (18.6)	-27.4 (47.8)	59,330
Different-type history (0–60m)	-13.2 (13.0)	-4.4 (3.0)	-0.3 (10.1)	44.0*** (13.8)	-7.8 (50.3)	59,330
Climate-related history (0–60m)	-9.5 (12.4)	-5.3* (3.1)	-4.6 (10.4)	48.5*** (15.6)	-10.0 (49.6)	59,330
Prior climate episodes	-24.9 (23.9)	-6.7 (5.1)	-2.0 (13.7)	42.3** (16.5)	-1.1 (53.7)	59,330
Panel D. Population and density						
Population	2.3 (15.9)	1.4 (3.6)	-16.5 (10.3)	39.9*** (13.7)	-16.1 (46.3)	59,330
Population density	-10.1 (19.1)	0.2 (4.0)	-10.9 (12.7)	31.1** (15.6)	-9.1 (43.9)	59,330

Notes: Each row reports a separate regression of the fixed-price change in the county green-ETF portfolio share, in basis points. Columns (1)–(3) report the characteristic's interactions with the direct-hit indicator, climate framing, and their interaction. The Surviving Hit × Framing column is Hit × Framing from the regression with the full characteristic block—Hit × M, Framing × M, Hit × Framing × M, and the characteristic level where not absorbed—added, evaluated at the estimation-sample characteristic means (sample category shares for indicators); the adjacent Affected column comes from that same regression. Framing and log total disaster-news attention both use event weeks 0–3. Models include the baseline controls and county, quarter, and episode fixed effects, and cluster standard errors by county and episode. Continuous characteristics are centered and divided by their estimation-sample standard deviation; indicators are not scaled; the no-history indicator is uncentered, so its surviving slope is evaluated at the history-present reference group. Perceived scientific consensus uses the last pre-2018 county value; the CDC vaccination subproxy is unavailable. Policy talk is the local-news share discussing climate policy or green issues, measured contemporaneously or one quarter earlier. News penetration is local-outlet circulation per county resident. Pre-2018 ETF turnover measures gross portfolio reallocation across all ETF positions: absolute position changes divided by lagged assets, averaged within county before 2018. Counties are split into terciles, with low omitted and middle controlled; the share of active quarters was degenerate, so the table uses ruled candidate (b). Own-county damage is zero by design for every non-hit county, so its separate framing interaction is structurally unidentified and shown as a dash; its remaining cells and surviving effect come from one regression that also controls for episode damage. Disaster-history counts enter in logarithms, with same-type history centered among counties with such history. The sample covers flood, hurricane and coastal episodes, 2018Q1–2024Q4. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

TABLE F.2: EARTHQUAKE PLACEBO: LOCAL GREEN-ETF SHARE AND CLIMATE FRAMING

	Dependent variable:	
	Fixed-price green-ETF share change (bp)	
Interpretation	Event quarter	Mean through $q-1$
	(1)	(2)
Hit × Interpretation (bp per SD)	1.01	−.86
	(4.53)	(7.26)
Hit	10.00	10.42
	(6.27)	(7.12)
Interpretation	−1.56	5.01
	(2.66)	(15.42)
Coverage	3.64	3.59
	(6.33)	(5.26)
Hit × Coverage	3.11	3.51
	(7.83)	(6.68)
p-value: Hit × Interpretation	.824	.906
Observations	180,934	195,425
Treated counties	17	18
Treated county–quarters	32	34
Treated episodes	52	54
County, quarter and episode FE	✓	✓

Notes. This table reports the county green-ETF share response to earthquakes (USGS ComCat). The outcome is the fixed-price share in quarter $q+1$ minus the share in quarter $q-1$, q the earthquake quarter, in basis points. Interpretation is the county's climate interpretation of other disasters, in the event quarter (column 1) or averaged through $q-1$ (column 2, at least four earlier observations), standardized within sample; Coverage is standardized event-quarter coverage. County, calendar-quarter and earthquake-episode fixed effects. Episodes with a large climate disaster in a directly hit county in $q-1$ to $q+1$ are excluded. Standard errors, in parentheses, are clustered by county. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE F.3: DISASTER NEWS AND TURNOVER OF ETF AND TOTAL REPORTED PORTFOLIOS

	Outcome: log(1 + turnover in bp), quarter $t+1$	
	(1) ETFs	(2) All reported securities
Affected × Interpretation	0.050	-0.003
	(0.065)	(0.055)
Non-affected × Interpretation	0.028	0.032**
	(0.018)	(0.014)
Affected × Attention	0.121*	0.107**
	(0.062)	(0.054)
Non-affected × Attention	0.091***	0.081***
	(0.030)	(0.022)
Interpretation: affected minus non-affected	0.023	-0.035
	(0.067)	(0.055)
Attention: affected minus non-affected	0.031	0.026
	(0.062)	(0.053)
Observations	101,785	101,785
R^2	0.135	0.188
Affected county-episodes	236	236
Episode and quarter FE	✓	✓
State FE	✓	✓
Affected × hazard × year FE	✓	✓
Financial and demographic controls	✓	✓
Own-county damage, deaths and injuries	✓	✓
Political and belief controls	✓	✓
Lagged climate attention	✓	✓
Affected × controls	✓	✓

Notes. Affected and non-affected slopes are estimated jointly in each column. We adapt Calvet, Campbell and Sodini (2009, online Appendix, Section 6.1) to quarterly adviser holdings. For adviser i and book B, $T_{iB,t+1} = \sum_{j \in B} \widetilde{P}_{j,t} |Q_{ij,t+1} - Q_{ij,t}| / A_{iB,t}$, where $A_{iB,t} = \sum_{j \in B} MV_{ij,t}$ is starting reported book value. County turnover is $T_{cB,t+1} = \sum_{i \in c} A_{iB,t} T_{iB,t+1} / \sum_{i \in c} A_{iB,t}$. Absolute changes are computed within each adviser before county aggregation; opposing adviser trades both count. No factor of one-half is applied. The outcome is $\log(1 + 10^4 T_{cB,t+1})$, with the log applied after county aggregation. Logging follows the log(1 + turnover) approach in Anderson (2013); basis-point scaling is specific to this application. Both columns use the same advisers with adjacent valid reports, unchanged county assignments and identical county-episode estimation rows. Column (1) covers ETFs; column (2) covers all reported security classes. Prices are security-quarter median unit values at t, with $t+1$ values used when unavailable; pairs with any unpriced position are excluded. The horizon is the end of event quarter t to the end of $t+1$, without an additional clean-window restriction. Quarterly snapshots omit trades reversed within the quarter; holdings changes may reflect investor inflows or outflows. The event universe is high-salience flood, hurricane and coastal episodes in 2015–2024. Attention and Interpretation use the first four weeks and are standardized. Controls and fixed effects follow Table 2, column (4). Difference-row standard errors use the full joint covariance. Standard errors are clustered by county. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE F.4: LOCAL CLIMATE INTERPRETATION AND ADVISER PORTFOLIOS NEAR DISASTERS

	Dependent variable:				
	Green-ETF share response (bp) Pre-event prices				
	(1)	(2)	(3)	(4)	(5)
Panel A. $h = 0$					
Interpretation effect among:					
Affected or nearby advisers, 0–50 miles	7.97***	8.21**	6.69***	6.64***	7.32**
	(2.75)	(3.18)	(2.44)	(2.53)	(3.09)
Observations	6,481	6,481	6,481	6,481	6,481
R^2	.069	.074	.100	.101	.272
Advisers	1,030	1,030	1,030	1,030	1,030
ZIP codes	431	431	431	431	431
Counties	124	124	124	124	124
Episodes	303	303	303	303	303
Panel B. $h = 1$					
Interpretation effect among:					
Affected or nearby advisers, 0–50 miles	11.75**	11.60**	11.59**	10.70**	13.12*
	(4.57)	(5.79)	(5.42)	(4.99)	(7.85)
Observations	5,889	5,889	5,889	5,889	5,889
R^2	.181	.185	.215	.218	.454
Advisers	992	992	992	992	992
ZIP codes	416	416	416	416	416
Counties	123	123	123	123	123
Episodes	295	295	295	295	295
Episode × state FE	✓	✓	✓		
Episode × band FE			✓		
Episode × state × band FE				✓	✓
County FE	✓	✓	✓	✓	
Adviser FE					✓
Attention controls	✓	✓	✓	✓	✓
Financial and demographic controls	✓	✓	✓	✓	✓
Own-county damage, deaths and injuries	✓	✓	✓	✓	✓
Political controls	✓	✓	✓	✓	✓
Lagged climate attention	✓	✓	✓	✓	✓
Group × controls		✓	✓	✓	✓

Notes. This table reports within-episode regressions of green-ETF portfolio changes among advisers in affected counties and nearby advisers within 50 miles of the affected area. The outcome is the share change from the quarter before the episode to the disaster quarter (Panel A) or the following quarter (Panel B), in basis points, with holdings valued at pre-event prices. Interpretation is the climate-interpreted share of disaster coverage at the adviser's ZIP over the episode's first four weeks. Coefficients are reported per ten-percentage-point increase and are common across affected and nearby advisers. All columns control for log disaster-news volume and its interaction with Affected; lagged log ETF and total Form 13F holdings; county income, college share, white share, population, density and prior Democratic vote share; own-county damage, deaths and injuries; and lagged climate-policy news share. Group × controls allows financial, demographic and political coefficients to differ across affected, nearby 0–25 and nearby 25–50 mile bands, which also define the band fixed effects. Observations are weighted by advisers' shares of the pre-event ETF book within each ZIP × episode. Standard errors, in parentheses, are clustered by physical shock and adviser-office ZIP. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE F.5: WHOSE VIEWS: COUNTY VIEWS AGAINST THE ADVISERS' OWN LEAN

	Dependent variable: ETF green-share response, $h = 1$ (bp)					
	ETF-book-share weights				Equal weights	
	Adviser lean	County vote	Both	Both, giving	Both	Both, giving
	(1)	(2)	(3)	(4)	(5)	(6)
Interpretation response, at hit-row means	19.40***	21.55***	21.65***	21.34***	5.44	5.25
	(7.43)	(7.96)	(8.03)	(7.98)	(6.54)	(5.97)
Disaster response, at hit-row means	5.91	13.24**	13.77**	8.12*	0.43	−2.58
	(4.74)	(5.57)	(5.92)	(4.71)	(5.67)	(5.00)
Interpretation response × County Rep. vote		6.90*	10.64*		10.52**	
		(3.96)	(5.48)		(5.15)	
Interpretation response × Adviser Rep. lean	−8.37		−13.84	−5.08	−7.82	−6.34
	(8.33)		(10.31)	(9.56)	(8.62)	(8.09)
Disaster response × County Rep. vote		7.54	6.56		3.41	
		(5.88)	(5.97)		(3.85)	
Disaster response × Adviser Rep. lean	1.52		0.79	−3.01	−9.83*	−10.71**
	(5.17)		(5.33)	(5.12)	(5.20)	(4.88)
Interpretation: adviser minus county			−24.48*	−2.49	−18.34	−12.58
			(14.83)	(12.73)	(12.92)	(10.66)
Interpretation response × County Rep. giving				−2.60		6.25*
				(5.01)		(3.29)
Disaster response × County Rep. giving				11.49*		5.45**
				(6.40)		(2.69)
Observations			465,705			
Hit rows (with an adviser lean)			1158 (723)			
Advisers			2130			

Interpretation response by county political group (slope within group, group-specific disaster response removed)

	ETF-book-share weights			Equal weights		
	Rep. share < .33	.33–.50	≥ .50	Rep. share < .33	.33–.50	≥ .50
Interpretation response	9.21	41.29**	21.47**	−4.42	17.24	5.35
	(6.51)	(16.71)	(8.36)	(5.33)	(14.08)	(5.17)

Adviser-by-event rows, flood, hurricane and coastal episodes, 2015Q1–2024Q4; episode and calendar-quarter fixed effects, Hit × year, attention and its interactions, prior green change, lagged portfolio size and the no-lean indicator included; standard errors clustered by county. Interpretation and attention per SD. Adviser Republican lean: the roster people's own political giving where a match exists, the firm's investment-decision staff giving otherwise; zero is a politically neutral firm. County Republican vote: two-party presidential share, mean over the sample elections, per SD across the sample counties; county giving tilt likewise. The response rows are evaluated at the hit-row means of all moderators. Book-share weights weight each adviser by its share of the county's ETF book; equal weights count each adviser-event link once. The interpretation entries report affected-minus-comparison differences in slopes. The affected-county responses discussed in Online Appendix OA.D add the corresponding comparison-group slopes; their standard errors use the full covariance matrix. The by-group panel re-estimates the model with group-specific disaster responses and interpretation slopes (group cutoffs on the county Republican vote share); equality of the three slopes is not rejected (book-share $p = .145$, equal $p = .117$). * $p < .10$, ** $p < .05$, *** $p < .01$.

G Appendix to Section 4

TABLE G.1: LOCAL HEAT AND THE ENVIRONMENTAL-PILLAR TILT OF MUTUAL FUNDS

	Dependent variable:	
	Environmental-Pillar tilt response (basis points)	
	Pre-event prices	Current prices
	(1)	(2)
Heat exposure	65.39*	71.40**
	(37.74)	(34.46)
Observations (fund-quarters)	53,802	53,802
Funds	2,335	2,335
Location-weighted county effects	✓	✓
Fund style × quarter FE	✓	✓
Fund controls	✓	✓
Heat severity controls	✓	✓

Notes. This table reports regressions of the quarterly change in the fund's Environmental-Pillar tilt on heat exposure, the share of the fund's offices in counties with a heat wave causing injuries or deaths in the quarter (SHELDUS), on fund-quarters from 2010 to 2023. The tilt is the weight of top- minus bottom-tercile stocks, in basis points, at pre-event prices (column 1) and current prices (column 2). Fund controls are the past-year return and flow, expense ratio, turnover, log assets and log age; severity controls are the deaths, injuries and damages of the heat events. Standard errors, clustered by fund and quarter, are in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE G.2: MUTUAL FUND PORTFOLIO RESPONSE: ALL FUND-EPISODES AND FUND-EPISODES WITH POLITICAL MEASURES

	Dependent variable: response in basis points									
	Pre-event prices					Current prices				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A. Fund level: change in the fund's Environmental-Pillar tilt										
Interpretation	50.25***	48.15***	69.34***	69.38***	52.03**	42.47***	39.76**	71.03***	71.49***	53.39**
	(12.80)	(12.51)	(20.11)	(19.87)	(20.69)	(15.46)	(15.37)	(20.85)	(21.85)	(22.64)
Attention	−1.06	−1.06	−2.05	−.85	−.15	−2.27	−2.32	−4.52	−3.79	−1.91
	(2.39)	(2.54)	(2.84)	(3.12)	(2.27)	(2.34)	(2.44)	(2.98)	(3.36)	(2.32)
Observations (fund-episodes)	51,226	51,226	37,047	37,047	37,051	51,226	51,226	37,047	37,047	37,051
Nearby fund-episodes	2,280	2,280	1,712	1,712	1,712	2,280	2,280	1,712	1,712	1,712
R^2	.025	.025	.031	.032	.025	.039	.039	.044	.045	.037
Exposure × episode FE	✓	✓	✓	✓		✓	✓	✓	✓	
Episode FE					✓					✓
Exposure × hazard × quarter FE					✓					✓
County FE	✓	✓	✓	✓		✓	✓	✓	✓	
Fund style FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Fund controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Damage and coverage controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Exposure × county controls		✓		✓	✓		✓		✓	✓
Sample with political measures			✓	✓	✓			✓	✓	✓
Panel B. Stock level: change in the fund's weight in a stock, High-E minus Low-E										
Interpretation		4.44***		3.57***			4.51***		3.66***	
		(1.28)		(1.28)			(1.21)		(1.33)	
Attention		−.09		.04			−.16		−.09	
		(.18)		(.27)			(.19)		(.29)	
Observations (fund-episode-stocks)		7,125,159		5,426,327			7,125,159		5,426,327	
Fund-episodes		72,461		53,028			72,461		53,028	
R^2		.539		.577			.550		.586	
Fund × episode FE		✓		✓			✓		✓	
Fund style × stock × episode FE		✓		✓			✓		✓	
Exposure × episode FE		✓		✓			✓		✓	
Damage and coverage controls		✓		✓			✓		✓	
Sample with political measures				✓					✓	

Notes. See notes to Table 3. Columns 1, 2, 6 and 7 use all fund-episodes, including managers and management companies without matched political donations; the other columns the fund-episodes with the political measures, as in Table 3. Columns 5 and 10 replace the exposure-by-episode and county fixed effects with episode fixed effects and fixed effects for each exposure band, hazard and calendar quarter.

TABLE G.3: MUTUAL FUND PORTFOLIO RESPONSE: MANAGER LOCATION AND CONTINUOUS SCORE

	Dependent variable: response in basis points									
	Pre-event prices					Current prices				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A. Funds with all offices in one county										
Interpretation	57.75***	53.93***	81.00***	79.28***	74.74***	55.02***	50.86***	76.93***	75.17***	70.43***
	(15.51)	(15.89)	(22.60)	(22.77)	(21.80)	(15.81)	(16.93)	(22.04)	(23.18)	(22.37)
Attention	−1.61	−1.59	−2.58	−1.52	−2.41	−2.99	−3.05	−4.27	−3.69	−4.64
	(2.79)	(2.82)	(3.26)	(3.49)	(3.43)	(2.83)	(2.83)	(3.40)	(3.73)	(3.56)
Observations (fund-episodes)	37,590	37,590	25,559	25,559	25,559	37,590	37,590	25,559	25,559	25,559
Nearby fund-episodes	2,061	2,061	1,404	1,404	1,404	2,061	2,061	1,404	1,404	1,404
Panel B. Fund located at all its offices										
Interpretation	53.10***	49.74***	72.30***	71.79***	68.23***	47.75***	43.91***	72.26***	71.63***	68.45***
	(13.81)	(14.01)	(20.06)	(19.75)	(19.01)	(14.25)	(14.63)	(21.16)	(21.84)	(21.23)
Attention	−2.77	−2.70	−4.42	−3.23	−3.58	−3.93	−3.83	−6.30*	−5.46	−5.77
	(2.90)	(3.03)	(3.13)	(3.46)	(3.46)	(2.98)	(3.01)	(3.34)	(3.64)	(3.58)
Observations (fund-episodes)	51,252	51,252	37,080	37,080	37,080	51,252	51,252	37,080	37,080	37,080
Nearby fund-episodes	2,531	2,531	1,884	1,884	1,884	2,531	2,531	1,884	1,884	1,884
Panel C. Fund located at its office closest to the disaster										
Interpretation	43.29***	38.62***	62.29***	63.50***	57.90***	45.16***	41.10***	65.08***	67.20***	61.61***
	(14.02)	(13.95)	(17.88)	(17.39)	(16.69)	(14.01)	(13.83)	(18.70)	(19.15)	(18.48)
Attention	−2.42	−2.03	−2.54	−1.21	−1.86	−3.99	−3.44	−4.79*	−3.44	−4.10
	(2.43)	(2.58)	(2.70)	(3.10)	(3.04)	(2.48)	(2.56)	(2.82)	(3.13)	(3.03)
Observations (fund-episodes)	51,257	51,257	37,084	37,084	37,084	51,257	51,257	37,084	37,084	37,084
Nearby fund-episodes	2,389	2,389	1,784	1,784	1,784	2,389	2,389	1,784	1,784	1,784
Panel D. Continuous Environmental-Pillar score										
Interpretation	14.64***	13.47***	18.54**	19.37**	17.66**	13.55**	12.03**	20.48**	21.30**	19.54**
	(4.65)	(4.73)	(7.73)	(8.16)	(8.19)	(5.91)	(5.82)	(8.92)	(9.33)	(9.44)
Attention	−.18	−.24	−.28	.09	−.12	−.35	−.39	−.92	−.60	−.79
	(.86)	(1.04)	(1.12)	(1.40)	(1.49)	(.91)	(1.07)	(1.17)	(1.47)	(1.54)
Observations (fund-episodes)	51,226	51,226	37,047	37,047	37,047	51,226	51,226	37,047	37,047	37,047
Nearby fund-episodes	2,280	2,280	1,712	1,712	1,712	2,280	2,280	1,712	1,712	1,712
Exposure × episode FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Fund style FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Fund controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Damage and coverage controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Exposure × county controls		✓		✓	✓		✓		✓	✓
Exposure × political controls					✓					✓
Sample with political measures			✓	✓	✓			✓	✓	✓

Notes. See notes to Table 3. Each panel changes one feature of the specification in Table 3. Panel A keeps the funds with all their offices in one county. Panels B and C place the fund at all its offices, with equal weight, or only at its office closest to the disaster. In Panel D the dependent variable sums, over the fund's scored stocks, the change in each stock's weight times the difference between its Environmental-Pillar score, from 0 to 1, and the fund's pre-event average score. Columns 1, 2, 6 and 7 use all fund-episodes; the other columns the fund-episodes with the political measures.

TABLE G.4: MUTUAL FUND PORTFOLIO RESPONSE BEFORE AND AFTER THE EPISODE

	Dependent variable:					
	Environmental-Pillar tilt response (basis points)					
	Pre-event prices			Current prices		
	(1)	(2)	(3)	(4)	(5)	(6)
Interpretation × Pre (−4, −3)	3.04	13.86	11.83	15.38	22.21	19.82
	(16.93)	(19.28)	(19.43)	(15.00)	(18.93)	(18.71)
Interpretation × Post (0 to 2)	70.28***	71.03***	67.50***	61.26***	69.56***	68.29***
	(20.04)	(20.56)	(19.72)	(20.25)	(21.88)	(21.35)
Observations (fund-episode quarters)	251,856	251,856	251,856	251,856	251,856	251,856
Fund-episodes	37,066	37,066	37,066	37,066	37,066	37,066
R^2	.375	.375	.376	.384	.384	.384
Fund-episode FE	✓	✓	✓	✓	✓	✓
Exposure × episode FE, by bin	✓	✓	✓	✓	✓	✓
County and fund style FE, by bin	✓	✓	✓	✓	✓	✓
Fund controls, by bin	✓	✓	✓	✓	✓	✓
Exposure × county controls, by bin		✓	✓		✓	✓
Exposure × political controls, by bin			✓			✓

Notes. See notes to Table 3. This table reports the regression of Table 3, Panel A, on quarters −4 to 2 around each episode. The dependent variable is the change in the tilt since the end of quarter −1; every regressor is interacted with Pre (quarters −4 and −3) and Post (quarters 0 to 2), with quarters −2 and −1 as the base.

TABLE G.5: MUTUAL FUND PORTFOLIO RESPONSE: AFFECTED AND DISTANT MANAGERS

	Dependent variable:					
	Environmental-Pillar tilt response (basis points)					
	Pre-event prices			Current prices		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Affected managers						
Interpretation	75.15**	73.98**	81.18***	70.18**	70.47**	77.54**
	(30.54)	(31.05)	(31.23)	(30.06)	(30.45)	(31.32)
Attention	−37.73***	−37.98***	−41.09**	−47.47***	−47.91***	−49.82***
	(12.99)	(13.05)	(17.05)	(13.01)	(13.28)	(16.99)
Fund-episodes in the band	1,973	1,973	1,961	1,973	1,973	1,961
Panel B. Managers more than 100 miles away						
Interpretation	3.70**	3.70**	3.34**	3.43**	3.45**	3.16*
	(1.52)	(1.52)	(1.65)	(1.71)	(1.70)	(1.83)
Attention	−.41	−.40	−.57	−.52	−.52	−.57
	(.58)	(.58)	(.60)	(.64)	(.64)	(.64)
Observations	37,047	37,047	36,964	37,047	37,047	36,964
R^2	.032	.033	.093	.045	.045	.102
Exposure × episode FE	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓
Fund style FE	✓	✓	✓	✓	✓	✓
Fund FE			✓			✓
Fund controls	✓	✓	✓	✓	✓	✓
Damage and coverage controls	✓	✓	✓	✓	✓	✓
Exposure × county controls	✓	✓	✓	✓	✓	✓
Exposure × political controls		✓	✓		✓	✓

Notes. See notes to Table 3. This table reports, from the regressions of Table 3, columns 2 to 4 and 6 to 8, the effects among managers with an office in the disaster zone (Panel A) and more than 100 miles from it (Panel B).

TABLE G.6: MUTUAL FUND ACTIVE TRADING

	Dependent variable:			
	Response (basis points of fund assets)			
	(1)	(2)	(3)	(4)
Interpretation	30.37***	30.14***	28.81***	29.80***
	(8.36)	(7.43)	(6.50)	(6.90)
Attention	.21	1.32	1.14	.68
	(1.36)	(1.52)	(1.48)	(1.63)
Nearby fund-episodes	1,806	1,806	1,806	1,804
Observations	38,177	38,177	38,177	38,091
R^2	.025	.026	.026	.087
Exposure × episode FE	✓	✓	✓	✓
County FE	✓	✓	✓	✓
Fund style FE	✓	✓	✓	✓
Fund FE				✓
Fund controls	✓	✓	✓	✓
Damage and coverage controls	✓	✓	✓	✓
Exposure × county controls		✓	✓	✓
Exposure × political controls			✓	✓

Notes. See notes to Table 3. The dependent variable is net active purchases of top- minus bottom-tercile stocks in the episode quarter, in basis points of lagged assets; as in Sammon and Shim [2023], the change in a continuing position is net of its proportional scaling with the fund's flow, and the whole change in a position opened or closed in the quarter counts as active.

TABLE G.7: DECOMPOSITION OF THE PORTFOLIO RESPONSE

	Dependent variable:							
	Environmental-Pillar tilt response component (basis points)							
	Within distance band		Across distance bands		Within sector		Across sectors	
	Pre-event	Current	Pre-event	Current	Pre-event	Current	Pre-event	Current
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Without political controls								
Interpretation	69.90***	71.60***	−.52	−.11	34.48*	31.73	34.90**	39.76**
	(19.49)	(22.06)	(4.74)	(4.53)	(17.66)	(20.02)	(14.98)	(15.86)
Attention	−1.62	−3.77	.77	−.02	−2.30	−3.61	1.45	−.18
	(3.21)	(3.34)	(.83)	(.82)	(2.78)	(3.00)	(2.12)	(2.25)
Nearby fund-episodes	1,712	1,712	1,712	1,712	1,712	1,712	1,712	1,712
Observations	37,047	37,047	37,047	37,047	37,047	37,047	37,047	37,047
R^2	.033	.043	.027	.029	.030	.038	.026	.030
Panel B. With political controls								
Interpretation	66.33***	68.76***	−.07	−.02	31.65**	28.54	34.60**	40.19**
	(18.44)	(21.20)	(4.65)	(4.74)	(15.97)	(18.30)	(14.85)	(15.81)
Attention	−1.88	−3.99	.78	−.02	−2.64	−4.01	1.55	.00
	(3.13)	(3.24)	(.80)	(.81)	(2.72)	(2.89)	(2.15)	(2.27)
Nearby fund-episodes	1,712	1,712	1,712	1,712	1,712	1,712	1,712	1,712
Observations	37,047	37,047	37,047	37,047	37,047	37,047	37,047	37,047
R^2	.033	.044	.027	.029	.031	.039	.027	.030
Exposure × episode FE	✓	✓	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓	✓	✓
Fund style FE	✓	✓	✓	✓	✓	✓	✓	✓
Fund controls	✓	✓	✓	✓	✓	✓	✓	✓
Damage and coverage controls	✓	✓	✓	✓	✓	✓	✓	✓
Exposure × county controls	✓	✓	✓	✓	✓	✓	✓	✓

Notes. See notes to Table 3. This table splits the dependent variable of Table 3, columns 2 and 6 (Panel A) and 3 and 7 (Panel B), into the changes within and across firm-distance bands and GICS sectors.

TABLE G.8: MUTUAL FUND RESPONSE ACROSS STOCKS: FIRM LOCATION

	Dependent variable:			
	Stock weight response, High-E minus Low-E (basis points)			
	Pre-event prices		Current prices	
	(1)	(2)	(3)	(4)
Panel A. Baseline				
Interpretation	3.57***	3.56***	3.66***	3.71***
	(1.28)	(1.28)	(1.33)	(1.29)
Observations	5,426,327	5,426,327	5,426,327	5,426,327
Fund-episodes	53,028	53,028	53,028	53,028
Fund × episode FE	✓	✓	✓	✓
Fund style × stock × episode FE	✓	✓	✓	✓
Exposure × political controls		✓		✓
Panel B. Firm in the disaster zone				
Interpretation, firms outside the zone	3.57***	3.54***	3.66***	3.69***
	(1.11)	(1.09)	(1.12)	(1.07)
Interpretation, firms in the zone	18.61**	18.52**	17.47**	17.34**
	(7.48)	(7.50)	(7.50)	(7.50)
Observations	5,272,599	5,272,599	5,272,599	5,272,599
Fund-episodes	53,028	53,028	53,028	53,028
Fund × episode FE	✓	✓	✓	✓
Fund style × stock × episode FE	✓	✓	✓	✓
Exposure × political controls		✓		✓
Panel C. Firm within 100 miles of the disaster zone				
Interpretation, firms more than 100 miles away	3.50***	3.46***	3.55***	3.57***
	(1.16)	(1.13)	(1.15)	(1.08)
Interpretation, firms within 100 miles	6.44**	6.48**	6.48**	6.62**
	(2.60)	(2.64)	(3.25)	(3.32)
Observations	5,272,599	5,272,599	5,272,599	5,272,599
Fund-episodes	53,028	53,028	53,028	53,028
Fund × episode FE	✓	✓	✓	✓
Fund style × stock × episode FE	✓	✓	✓	✓
Exposure × political controls		✓		✓

Notes. See notes to Table 3, Panel B. In Panels B and C every term is interacted with an indicator for the location of the firm's headquarters: in the disaster zone (Panel B) or within 100 miles of it (Panel C).

TABLE G.9: MANAGERS OR HEADQUARTERS: WHO RESPONDS

	Dependent variable:	
	Environmental-Pillar tilt response, pre-event prices (basis points)	
	(1)	(2)
Nearby managers, headquarters within 100 miles	66.21***	65.87***
	(21.00)	(20.72)
Nearby managers, headquarters beyond 100 miles	71.19***	69.07***
	(20.73)	(21.62)
Headquarters nearby, managers beyond 100 miles	18.74	21.04
	(32.42)	(33.06)
Managers nearby (headquarters far) minus headquarters nearby (managers far)	52.45*	48.04
	(30.64)	(33.80)
Fund-episodes (episodes), managers nearby, headquarters far	418 (46)	418 (46)
Fund-episodes (episodes), headquarters nearby, managers far	482 (75)	482 (75)
Observations	36,165	36,165
Exposure × episode FE	✓	✓
County FE	✓	✓
Fund style FE	✓	✓
Fund controls	✓	✓
Damage and coverage controls	✓	✓
Exposure × county controls		✓

Notes. See notes to Table 3. This table reports, in one regression with the specification of Table 3, columns 1 and 2, the interpretation effect among nearby managers separately for funds whose headquarters is within and more than 100 miles from the episode, and the effect of the headquarters county's interpretation for funds with no office within 100 miles but the headquarters within 50 miles. The sample is the fund-episodes whose headquarters location is known.

H Appendix to Section 5

TABLE H.1: OUTLET-LEVEL FIRST STAGE OF NEWSPAPER-CHAIN SIZE

	Dependent variable:	
	Disaster coverage (1)	Interpretation share (2)
Log chain size	−0.061	−0.023***
	(0.041)	(0.005)
Observations	6,711	6,711
R^2	.650	.362
First-stage F	2.20	20.85
Newspaper and quarter FE	✓	✓
Disaster coverage control		✓

Notes. This table reports newspaper–quarter regressions of disaster coverage and of the interpretation share on log chain size, for chained newspapers with positive disaster coverage, 2015–2024 (263 newspapers, 40 quarters). Disaster coverage is $\log(1 + A)$, A the number of disaster articles in the quarter; the interpretation share is the climate-interpreted share of those articles; log chain size is $\log(1 + S)$, S the number of other papers of the same owner, under the owner in place before the quarter. Column 2 controls for disaster coverage. Standard errors clustered by newspaper. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE H.2: CHAIN SIZE AND PREDETERMINED CHARACTERISTICS

	Dependent variable: chain size
Politics and beliefs	
Democratic two-party share, latest presidential election	0.057
	(0.257)
Share worried about global warming	-0.214
	(0.159)
Investors	
Green-ETF share change, two pre-event quarters	0.001
	(0.005)
Portfolio assets (log)	-0.033
	(0.028)
County	
Population growth, four years	0.024
	(0.052)
Income growth, four years	0.066
	(0.056)
Disaster	
Deaths in the county (log)	0.013
	(0.013)
Damage in the county (log)	-0.022
	(0.030)
Deaths outside the county (log)	-0.002
	(0.013)
Damage outside the county (log)	0.003
	(0.017)
Counties in the footprint (log)	0.007
	(0.014)
Distance to the disaster (miles)	0.022
	(0.017)
Local disaster coverage (log)	-0.035
	(0.025)
Audience overlap with the footprint	0.005
	(0.017)
Joint test, p-value	0.66
County-events	694
Counties	184

Notes. This table reports, for each characteristic, the coefficient from a regression of chain size on that characteristic, both standardized, on the county-event sample of Table 4, with the fixed effects of that table (county, calendar quarter, hazard by distance band). The joint test is from the regression of chain size on all characteristics. Standard errors clustered by county in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

I Appendix to Section 6

TABLE I.1: COUNTY GREEN-ETF PORTFOLIO RESPONSE BY EXPOSURE BAND

	Dependent variable:			
	Green-ETF share response (basis points)			
	Pre-event prices		Current prices	
	(1)	(2)	(3)	(4)
Affected × Interpretation		37.27***		37.93***
		(13.64)		(13.56)
Band × Interpretation	4.34	−9.54	4.30	−9.87
	(6.52)	(7.38)	(6.57)	(7.53)
Affected × Attention		−3.51		−4.70
		(14.10)		(13.99)
Band × Attention	−10.13*	−9.47	−9.31*	−8.09
	(5.55)	(7.81)	(5.27)	(7.41)
Total Interpretation effect among affected and nearby counties	4.35		4.32	
	(6.51)		(6.57)	
Total Interpretation effect among nearby counties		−9.52		−9.84
		(7.27)		(7.45)
Total Interpretation effect among affected counties		27.74**		28.08**
		(11.88)		(11.80)
Observations	105,662	105,662	105,662	105,662
R^2	.101	.102	.106	.106
Affected county-episodes	244	244	244	244
Nearby county-episodes	569	569	569	569
Affected and its interactions		✓		✓
Episode and quarter FE	✓	✓	✓	✓
Band × hazard × year FE	✓	✓	✓	✓
County FE	✓	✓	✓	✓
Controls and their band interactions	✓	✓	✓	✓

Notes. See notes to Table 2. Band is one for counties in the episode footprint or nearby, within 50 miles of it; distant counties form the comparison group. Columns 1 and 3 include the band indicator and its interactions with Interpretation and Attention only, so that the band rows compare affected and nearby counties together with distant counties. Columns 2 and 4 add Affected and its interactions, so that the Band rows compare nearby with distant counties and the Affected × Interpretation coefficient is the difference between the total Interpretation effects among affected and among nearby counties. Each regression also includes Interpretation, Attention, the band indicator and, in columns 2 and 4, Affected; their coefficients are not reported. Band, hazard and year effects replace the Affected, hazard and year effects; the controls of Table 2, column 5, enter with their band interactions, together with county fixed effects. The total Interpretation effect among a group is the sum of the Interpretation coefficient and its interactions for that group, with standard errors from their joint covariance. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE I.2: PERSISTENCE OF THE COUNTY GREEN-ETF PORTFOLIO RESPONSE

	Dependent variable: Green-ETF share response (basis points)						
Quarter	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$
Panel A. Pre-event prices							
Affected × Interpretation	19.64**	29.56**	16.30	48.33**	52.37*	140.54***	162.69**
	(8.57)	(11.67)	(42.41)	(23.15)	(31.08)	(46.06)	(64.59)
Total Interpretation effect among affected counties	20.29**	30.64***	18.05	48.45**	52.34*	139.52***	159.43**
	(8.62)	(11.34)	(41.68)	(23.30)	(31.31)	(46.35)	(64.63)
Observations	105,567	97,524	89,450	81,682	75,123	69,407	62,850
Affected county-episodes	244	224	195	179	151	139	121
County clusters	301	297	286	276	267	262	248
Panel B. Current prices							
Affected × Interpretation	19.39**	28.35**	15.20	48.79**	50.62*	132.18***	145.47**
	(8.68)	(11.55)	(42.81)	(23.56)	(28.88)	(43.58)	(57.78)
Total Interpretation effect among affected counties	20.04**	29.29***	16.77	48.61**	50.19*	130.60***	141.67**
	(8.73)	(11.24)	(42.14)	(23.68)	(29.05)	(43.87)	(57.91)
Observations	105,567	97,524	89,450	81,682	75,123	69,407	62,850
Affected county-episodes	244	224	195	179	151	139	121
County clusters	301	297	286	276	267	262	248

Notes. Each column is estimated separately. The outcome is the county green-ETF share at the end of quarter h minus its available mean over quarters -2 and -1, in basis points; $h = 0$ is the disaster quarter. Panel A values holdings at prices and green classifications from the quarter before the episode; Panel B uses current prices and classifications. Both panels use the same finite portfolio observations. The sample covers high-salience flood, hurricane and coastal episodes, 2015–2024, and excludes qualifying shocks in quarters -2 and -1 and subsequent shocks through h. The eligible sample varies across horizons. Interpretation and Attention, controls and fixed effects follow Table 2, columns (4) and (9). Fixed effects are state, quarter, episode, and Affected × hazard × year. Entries are responses to a one-standard-deviation increase in Interpretation (15.7 percentage points), conditional on Attention. Affected × Interpretation is the slope difference between affected and unaffected counties. The total effect among affected counties is the sum of the Interpretation coefficient and this interaction, with its standard error computed from their full covariance. Standard errors, in parentheses, are clustered by county. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE I.3: CLIMATE BELIEFS AND GREEN ETF HOLDINGS

	Dependent variable: Green share of ETF holdings (basis points)					
	Worried	U.S. harm now/within 10 years	U.S. harm already happening	Warming happening	Mostly human caused	Support CO_2 regulation
	(1)	(2)	(3)	(4)	(5)	(6)
Belief share	1.19***	1.09**	1.69**	1.32**	0.86	0.94
	(0.41)	(0.44)	(0.74)	(0.54)	(0.64)	(0.84)
DMA-years	128	128	128	135	135	126
Media markets	20	20	20	21	21	19
Advisers	1,227	1,227	1,227	1,251	1,251	1,210
Survey respondents	7,047	7,017	7,017	7,547	7,542	6,851
Mean green share (bp)	5.61	5.61	5.61	5.32	5.32	5.66
Year fixed effects	✓	✓	✓	✓	✓	✓

Notes. Annual DMA-level regressions, 2009–2019. Each column relates the green share of advisers' ETF holdings to the indicated binary belief share. Coefficients are basis points per 10-percentage-point higher belief share. Advisers predominantly serve households and all their known U.S. offices lie within one DMA. Belief shares require at least 30 valid responses per DMA-year. Questions and coding are given in Figure 6. Standard errors clustered by DMA are in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

TABLE I.4: CLIMATE BELIEFS AFTER DISASTERS

	Dependent variable: Binary survey answer (percentage points)					
	Worried	U.S. harm now/within 10 years	U.S. harm already happening	Warming happening	Mostly human caused	Support CO_2 regulation
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Average response						
Post-disaster	6.83	13.44**	11.41**	10.42**	0.69	6.09
	(5.99)	(5.19)	(4.78)	(4.32)	(5.87)	(4.15)
Respondents	1,172	1,157	1,157	1,204	1,203	1,131
Media markets	62	61	61	63	63	60
Panel B: Responses by ideology						
Post × Liberal	4.82	6.17	4.04	7.43	-7.18	1.48
	(6.24)	(8.32)	(7.90)	(5.26)	(6.84)	(5.49)
Post × Moderate	6.23	16.43***	12.72**	13.34**	3.57	13.23**
	(7.66)	(6.15)	(5.87)	(5.60)	(7.90)	(5.01)
Post × Conservative	8.85	15.39**	13.51**	5.68	2.38	3.36
	(7.53)	(7.33)	(6.20)	(5.07)	(6.12)	(6.23)
Respondents	1,145	1,131	1,131	1,177	1,176	1,112
Media markets	62	61	61	63	63	60
Liberal respondents	319	318	318	328	328	310
Moderate respondents	434	427	427	441	441	422
Conservative respondents	392	386	386	408	407	380
DMA × survey-year FE	✓	✓	✓	✓	✓	✓
Interview year-month FE	✓	✓	✓	✓	✓	✓
Event fixed effects	✓	✓	✓	✓	✓	✓
Demographic/wording controls	✓	✓	✓	✓	✓	✓

Notes. CCAM respondents in affected DMAs interviewed from 28 days before through 27 days after flood onset, 2009–2019. Coefficients are percentage-point changes in the indicated binary answer. Panel B includes ideology indicators and separate post-disaster coefficients for liberals, moderates and conservatives. Controls are age, college education, gender, income and question wording. Questions and coding are given in Figure 6. Standard errors clustered by DMA are in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

TABLE I.5: GREEN SHARES, FLOODS AND CLIMATE BELIEFS

	Green share	Answer	
	Belief	Post	Post × Interpretation
	(1)	(2)	(3)
Worry	.731*	.135	.137*
	(.319)	(.089)	(.055)
	[.058]	[.398]	[.059]
Observations	117	1,671	1,635
Media markets	117	76	71
Harm sooner	1.122***	.294***	.151**
	(.341)	(.082)	(.054)
	[.007]	[.003]	[.034]
Observations	117	1,654	1,618
Media markets	117	76	71
Happening	.912*	.181	.061
	(.384)	(.085)	(.088)
	[.058]	[.147]	[.493]
Observations	117	1,316	1,280
Media markets	117	70	66
Human causation	.841*	−.016	.088
	(.467)	(.104)	(.062)
	[.074]	[.881]	[.475]
Observations	117	1,702	1,666
Media markets	117	77	73
CO_2 regulation	.706**	.114	.081
	(.271)	(.081)	(.059)
	[.041]	[.398]	[.475]
Observations	117	1,516	1,480
Media markets	117	68	63
Media market × survey-year FE		✓	✓
Interview year-month FE		✓	✓
Episode FE		✓	✓
Age, college, gender, income, wording		✓	✓

Notes. This table reports three regressions for each of five climate belief questions from the Yale–George Mason Climate Change in the American Mind surveys. Column (1) regresses the media market's average 2010–2017 green share of adviser holdings on its mean answer before 2018, one observation per market; the coefficient is in basis points of green share per between-market standard deviation of the mean answer. Columns (2) and (3) use respondents in affected media markets interviewed from 28 days before to 27 days after a flood starts, with answers standardized within survey wave. Post equals one on or after the start day. Column (3) interacts Post with central climate attribution in local disaster coverage accumulated from the event's start through the interview day: the attribution share within each outlet, averaged across covering outlets with circulation weights. One standard deviation is 14.54 percentage points, measured across finite roster event-day cells. The interaction is zero before the event starts; post-event interviews without coverage leave this fit. Each sample retains before and after observations within media market by survey-year and removes singleton fixed effects. Units and samples differ across columns. Regressions are unweighted. Standard errors, clustered by media market, in parentheses. Holm-adjusted p-values in brackets, separately over the five questions within each column; stars use these adjusted values. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

TABLE I.6: WILDFIRE AND FLOOD NEWS ATTRIBUTION AND FLOOD INSURANCE TAKE-UP

	Dependent variable: $CNew_{c,t_e+h} - CNew_{c,t_e-1}$			
	Wildfire		Flood	
	$h = 2$	$h = 3$	$h = 2$	$h = 3$
	(1)	(2)	(3)	(4)
Hazard hit	14.48**	21.71**	32.04***	40.88***
	(7.15)	(10.48)	(8.06)	(10.35)
Total hazard-news volume	6.32	10.02	4.38	5.21
	(12.57)	(15.59)	(5.79)	(7.14)
Climate-attributed hazard-news volume	−8.49	−13.97	−1.58	−1.13
	(9.85)	(12.12)	(4.36)	(5.68)
Strict climate-only news volume	4.76**	6.07*	3.35*	3.39
	(2.40)	(3.50)	(1.85)	(2.15)
Hazard hit × Total hazard-news volume	−17.26*	−15.14	−10.71	−17.47
	(9.79)	(13.61)	(11.11)	(13.62)
Hazard hit × Climate-attributed hazard-news volume	17.82**	18.18*	−.15	2.11
	(8.07)	(10.02)	(8.14)	(9.80)
Hazard hit × Strict climate-only news volume	−1.27	−8.78	−2.36	−2.70
	(5.94)	(9.90)	(4.60)	(5.82)
Observations	881	881	4,206	4,206
Counties	190	190	904	904
Hazard episodes	50	50	227	227
R^2	.846	.835	.662	.676
County FE	✓	✓	✓	✓
Hazard-episode FE	✓	✓	✓	✓
State × Year FE	✓	✓	✓	✓

Notes. This table reports the response of first-time flood-insurance take-up to the volume of wildfire and flood news and to the share of it that links the hazard to climate change. The outcome is the first-lag difference $CNew_{c,t_e+h} - CNew_{c,t_e-1}$, so $t - 1$ is the base period. The wildfire and flood columns are estimated separately. The unit is county by hazard episode and each sample begins in 2010. $CNew$ is cumulative county NFIP first-time voluntary policy issuance per 100,000 residents. Columns (1) and (3) report horizon $h = 2$; columns (2) and (4) report horizon $h = 3$. News volumes use county local-circulation weights and days 0–13 after hazard onset. Each news measure is standardized within its hazard-specific estimation sample after applying $\log(1 + x)$. Every column includes county, hazard-episode, and state-by-event-year fixed effects; standard errors are clustered by county. The wildfire sample excludes counties hit by a flood or coastal event from $t - 6$ through $t + 3$. The flood sample applies the same exclusion from $t - 6$ through $t + 3$, since flood exposure directly affects policy issuance. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

TABLE I.7: PAST GREEN RETURNS AND THE ADVISER RESPONSE

	Dependent variable: ETF green-share response, $h = 1$ (bp)		
	Base	Prior green return added	Prior green return × interpretation
	(1)	(2)	(3)
Interpretation response	19.34***	19.33***	20.70***
	(7.14)	(7.10)	(7.21)
Disaster response × prior green return		0.23	
		(6.03)	
Interpretation response × prior green return			−1.45
			(4.79)
Interpretation response, prior green return −1 SD			22.15***
			(8.38)
Interpretation response, prior green return +1 SD			19.25**
			(8.92)
Observations		465,705	
Hit rows		1,158	

Adviser-by-event rows, flood, hurricane and coastal episodes, 2015Q1–2024Q4. Prior green return: the return of green ETFs minus all non-green ETFs over the 52 weeks before the event week, equal-weighted across funds, one value per episode, per SD (6.1 percentage points). All columns include attention and its interaction with the disaster, Hit × year, lagged log portfolio size and the county Democratic vote share; episode and calendar-quarter fixed effects. Column (2) adds the prior green return interacted with the disaster. Column (3) interacts the prior green return with interpretation, the disaster and attention; its level is absorbed by the episode fixed effects, and the lower-order terms are not reported. Observations are weighted by the adviser's share of the county's ETF book at $t - 1$. Standard errors clustered by county in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

J Aggregation: Framework and Calibration

1. Aggregate ESG flows

The additional allocations $a^I_{i,e}$ derived in Appendix B generate demand for ESG funds and their underlying stocks. Consider the resulting purchase of X dollars of the ESG portfolio, financed by selling X dollars of the broad market portfolio, at initial prices.

First, we establish how much additional ESG investment this transaction generates. Let K be the fixed capital of the investor population considered and s its initial share in ESG funds. The other-fund portfolio in the individual problem is the non-ESG component of this broad portfolio. Because the broad market portfolio includes ESG funds with share s, selling it offsets sX of the ESG purchase. The net dollar addition to ESG holdings is therefore $T = (1 - s)X$. Relative to initial ESG assets $G_0 = sK$, the corresponding proportional flow is

$$f = \frac{T}{G_0}. \tag{J.1}$$

2. Induced relative demand for green stocks

To trace the consequences for stock prices, we now look through the fund portfolios to the stocks they hold. The ESG purchase and broad-portfolio sale generate different stock demands because the two portfolios assign different weights to each stock.

Following van der Beck (2026), I define green stocks as those overweighted by ESG-labelled funds relative to the aggregate fund portfolio. In particular, the representative ESG portfolio pools the holdings of ESG-labelled funds: w_n^{ESG} equals their total dollar holdings of stock n divided by their aggregate assets. The benchmark weight w_n^{all} is constructed in the same way across all funds, including ESG funds. Defining $\tau_n = w_n^{\text{ESG}} - w_n^{\text{all}}$, green stocks ($\tau_n > 0$) form group G and the remaining stocks form group N.

I assume that the reallocation is implemented through purchases and sales of the underlying stocks in these portfolio proportions. Buying X dollars of the ESG portfolio therefore means demanding Xw_n^{ESG} of each stock n, while financing the purchase by selling the broad market portfolio generates sales of Xw_n^{all}. Writing w_n^O for the stock weight in the other-fund portfolio gives $w_n^{\text{all}} = sw_n^{\text{ESG}} + (1 - s)w_n^O$. The individual-model reallocation of T dollars from other funds into ESG funds therefore induces the same stock demand as the $X = T/(1 - s)$ broad-market-financed purchase:

$$T\left(w_n^{\text{ESG}} - w_n^O\right) = X\left(w_n^{\text{ESG}} - w_n^{\text{all}}\right) = X\tau_n. \tag{J.2}$$

The reallocation thus generates net purchases of green stocks and net sales of underweighted stocks.

Let τ_G and τ_N denote the sums of the weight differences within each group. Aggregating the stock demands gives total net demand for green and nongreen stocks of $X\tau_G$ and $X\tau_N$, respectively. Dividing each by its initial total market capitalization, V_G or V_N, gives the corresponding proportional flow. Their difference is the relative flow into green versus nongreen stocks induced by the initial reallocation of X dollars from the broad market portfolio into the ESG portfolio:

$$f^{\text{rel}} = \frac{X\tau_G}{V_G} - \frac{X\tau_N}{V_N}. \tag{J.3}$$

3. Valuation and price response

Finally, the relative prices of green and nongreen stocks adjust as other investors accommodate the reallocation. Van der Beck's (2026) *ESG flow multiplier*, denoted here by κ_{ESG}, measures the equilibrium difference in green and nongreen stocks' dollar valuation changes per dollar reallocated from the broad market portfolio into the ESG portfolio. It combines the weight differences τ_n defined above with holders' willingness to trade as prices change, captured by ownership-weighted demand elasticities estimated using other institutions' dividend-induced trading as an instrument for price changes.[62]

Let ΔV_G and ΔV_N denote the resulting changes in the total market values of green and nongreen stocks. Holding fundamentals and other demand drivers fixed, the calibrated first-order valuation response to this X-dollar reallocation, accounting for other investors' accommodation, is

$$\Delta V_G - \Delta V_N = \kappa_{\text{ESG}} X. \tag{J.4}$$

To express the same response in terms of relative returns, I define the *relative-return multiplier*, M_{ESG}, as the change in the return difference between value-weighted green and nongreen portfolios, $R_G - R_N$, per unit of relative flow f^{rel}. It is obtained from the same stock-price responses by dividing each portfolio's dollar valuation change by its initial market capitalization and the resulting return difference by f^{rel}:

$$\Delta(R_G - R_N) = \frac{\Delta V_G}{V_G} - \frac{\Delta V_N}{V_N} = M_{\text{ESG}} f^{\text{rel}}. \tag{J.5}$$

For the calibration, I use van der Beck's (2026) approach with his December 2022 replication inputs, including the portfolio weights used above to define green and nongreen stocks and their induced demand. These inputs give $\kappa_{\text{ESG}} \simeq 0.72$ and imply $M_{\text{ESG}} \simeq 1.12$.

4. From the local allocation response to aggregate ESG demand

The calculation in Parts 1–3 takes the aggregate reallocation X as given. I now use the local allocation estimates to determine the additional reallocation generated by stronger climate interpretation, holding disaster attention fixed. The comparison keeps other demand drivers unchanged, so baseline allocations may differ across investors while interpretation contributes an additional component to aggregate demand.

4.1. The additional allocation The event-induced demand shift derived in (B.7) is $u_{i,e} = \eta_e \gamma_{i,e}$. Allow the portfolio parameters to differ across investors, and denote investor i's responsiveness to log odds at initial prices by $\chi_i(P^0)$. The same solution gives $\gamma_{i,e} = \gamma_i(A_{i,e}, I_{i,e}, Z_e) = \chi_i(P^0)\ell_i(A_{i,e}, I_{i,e}, Z_e)$. Superscripts 0 and 1 denote the baseline and counterfactual information configurations. Relative to $(A^0_{i,e}, I^0_{i,e}, Z^0_e)$, the additional desired share at (A, I, Z) is

$$\begin{aligned} a_{i,e}(A, I, Z) &\equiv u_{i,e}(A, I, Z) - u_{i,e}(A^0_{i,e}, I^0_{i,e}, Z^0_e) \\ &= \eta_e \left[\gamma_i(A, I, Z) - \gamma_i(A^0_{i,e}, I^0_{i,e}, Z^0_e)\right]. \end{aligned} \tag{J.6}$$

[62] Van der Beck (2026), *Flow-Driven ESG Returns*, Sections 5.A, 5.C and 6.A, and Appendix A.3.

The calibration changes interpretation while holding attention and the wider information environment fixed. Its individual allocation increment is therefore

$$\begin{aligned} a^I_{i,e} &= u_{i,e}(A_{i,e}, I^1_{i,e}, Z_e) - u_{i,e}(A_{i,e}, I^0_{i,e}, Z_e) \\ &= \eta_e \chi_i(P^0) \left[\ell_i(A_{i,e}, I^1_{i,e}, Z_e) - \ell_i(A_{i,e}, I^0_{i,e}, Z_e)\right]. \end{aligned} \tag{J.7}$$

This is the individual counterpart of the estimated additional allocation $\widehat{a}^I$ in (B.17).

4.2. Aggregation Let K_i be investor i's initial eligible capital, with total capital $K = \sum_i K_i$. Denote ESG-fund units by Q_i. At initial prices, a portfolio share q_i purchases $Q_i = K_i q_i / P^0$ units. Write $Q^0_{i,e}$ and $Q^1_{i,e}$ for baseline and counterfactual units, and $a_{i,e}$ for the corresponding share increment. They satisfy

$$P^0\big(Q^1_{i,e} - Q^0_{i,e}\big) = K_i a_{i,e}. \tag{J.8}$$

This converts the derived share increment into desired dollar demand at initial prices. Summing across investors gives

$$T = \sum_i P^0\big(Q^1_{i,e} - Q^0_{i,e}\big) = \sum_i K_i \, a_{i,e}(A^1_{i,e}, I^1_{i,e}, Z^1_e). \tag{J.9}$$

Additional ESG purchases are financed by reducing other holdings, keeping each investor's capital fixed.

For the interpretation calibration, let $\mathcal{A}$ contain the investors activated by the counterfactual. Their combined capital represents a share α of total eligible capital:

$$\alpha = \frac{\sum_{i \in \mathcal{A}} K_i}{K}. \tag{J.10}$$

The central assumption is that investors "activated" in my counterfactual analyses would respond to stronger climate interpretation as the disaster-exposed investors in my sample do, at the specified level of attention and information environment. I use $\widehat{a}^I$ to estimate their capital-weighted average additional allocation a^I per dollar of initial capital, while allowing individual responses, initial portfolios and baseline responses to differ. I apply this increment once to each investor's initial capital, treating it as the cumulative allocation adjustment over the event quarter and the following quarter.

The resulting additional ESG investment is

$$T^I = \sum_{i \in \mathcal{A}} K_i a^I_{i,e} = \alpha K a^I. \tag{J.11}$$

Using the accounting in Part 1, its proportional ESG flow and equivalent broad-market-to-ESG reallocation are

$$f^I = \frac{T^I}{G_0}, \qquad X^I = \frac{T^I}{1-s}, \qquad G_0 = sK, \tag{J.12}$$

where s is the target population's initial ESG share.

Setting $\alpha = 1$ extends the estimated interpretation increment to all eligible capital, at the specified attention and information environment. Interpreting this calculation as the complete response to a national change additionally requires that any demand generated by a change in Z_e be negligible or separately included.

To obtain the associated price response, I assume that the reallocation translates fully into underlying stock demand in the portfolio proportions defined above. **Assumption: market accommodation.** I hold

the estimated stock-demand elasticities and initial ownership weights fixed as the set of activated investors changes. Activation shifts desired allocations, while both activated and nonactivated investors retain their price responsiveness. These are the stock-quantity elasticities entering van der Beck's multiplier. Substituting X^I for X in Parts 2 and 3 gives the additional relative stock flow, denoted $f^{\text{rel},I}$, and the associated valuation changes and returns.

4.3. The national information environment In the calibration, I abstract from changes in the national information environment, holding Z_e fixed as I extend the estimated interpretation response to the selected population.

A nationally covered event can also change Z_e. Public coverage and framing can shape expectations about other investors' responses (Morris and Shin, 2002). Investors might consequently buy green assets to profit from the price increases they expect others' purchases to generate (Schmidt-Engelbertz and Vasudevan, 2025).[63] This speculative motive lies outside the climate-belief channel modeled here. Within the model, Z_e changes $\gamma_i(A, I, Z)$ through its effect on the climate-belief update $\ell_i(A, I, Z)$.

Let $T^{\text{fixed } Z}$ be the sum of investors' demand changes when attention and interpretation take their counterfactual values but the national information environment remains at Z_e^0. For the interpretation-only exercise, this is the calibrated T^I. With full activation ($\alpha = 1$), the counterfactual asks what would happen if every investor made the same additional reallocation in response to stronger climate interpretation as the disaster-exposed investors in my sample, at the specified level of attention. National publicity can additionally change the climate-belief update through Z_e.

Denote by T^Z the additional aggregate demand generated through climate-belief updating when that environment changes to Z_e^1, holding the same attention and interpretation profiles fixed:

$$\begin{aligned} T^Z &= \sum_i K_i \left[\gamma_i(A_{i,e}^1, I_{i,e}^1, Z_e^1) - \gamma_i(A_{i,e}^1, I_{i,e}^1, Z_e^0)\right] \eta_e, \\ T^{\text{national}} &= T^{\text{fixed } Z} + T^Z. \end{aligned} \tag{J.13}$$

The term T^Z, which is not included in my calibration, captures what the change in national coverage and framing adds beyond the specified changes in individual attention and interpretation. It can include both a response shared across investors and changes in how strongly they respond to attention or interpretation. When $T^Z > 0$, the national climate-belief response exceeds the sum of local responses evaluated at the baseline national information environment.

5. Calibration exercise

For this calibration exercise, I measure attention and interpretation using newspaper coverage of the event, assuming that coverage captures its broader local salience and framing captures the local discussion and narratives attributing it to climate change. I control for other important determinants of attention and interpretation, such as whether the county was affected and local political views, which can also influence interpretation and thus portfolio choices. By holding these factors at their average levels in the calibration sample, I focus on the additional event-specific portfolio response induced by variation in interpretation captured by local climate narratives.

[63] See Schmidt-Engelbertz and Vasudevan (2025), *Speculating on Higher-Order Beliefs*, *Review of Financial Studies* 38(8), 2434–2466. Using investor surveys, they document that beliefs about others' beliefs can motivate positions that conflict with investors' own market valuations.

5.1. Applying the estimated allocation response The numerical calibration uses the same interpretation contrast as (B.17), evaluated among affected counties. Its outcome is the cumulative change in the green-ETF share from the pre-event quarter through the following quarter, valuing holdings at pre-event prices. I apply the price-response assumption of Appendix B to this measured share and horizon. The impact response in that section and the following-quarter response used here are separate estimates of the corresponding allocation adjustments.

Let $\widehat{\beta}_I^{\text{cal}} + \widehat{\delta}_I^{\text{cal}}$ be the estimated interpretation effect among affected counties for this calibration outcome, reported in basis points per standard deviation. Holding coverage and the other covariates fixed, the additional allocation in decimal share units is

$$\widehat{a}^I = \frac{\widehat{\beta}_I^{\text{cal}} + \widehat{\delta}_I^{\text{cal}}}{10{,}000} \frac{I^1 - I^0}{\sigma_I}. \tag{J.14}$$

This is the allocation increment used in $T^I = \alpha K \widehat{a}^I$.

For coverage, let $\widehat{\beta}_A^{\text{cal}} + \widehat{\delta}_A^{\text{cal}}$ be the affected-county response in basis points per standard deviation of $\log(1 + A)$, and σ_A that standard deviation. For a joint change in coverage and interpretation, the estimated additional share $\widehat{a}$ is

$$\widehat{a} = \frac{1}{10{,}000} \left[(\widehat{\beta}_A^{\text{cal}} + \widehat{\delta}_A^{\text{cal}}) \frac{\log(1 + A^1) - \log(1 + A^0)}{\sigma_A} + (\widehat{\beta}_I^{\text{cal}} + \widehat{\delta}_I^{\text{cal}}) \frac{I^1 - I^0}{\sigma_I} \right], \tag{J.15}$$

Here A^0, A^1 and I^0, I^1 are the baseline and counterfactual news measures.

5.2. A stronger climate interpretation Consider an increase in climate framing from its 25th to its 75th percentile, holding coverage at its median. In the calibration sample, this raises the share of coverage attributing the event to climate change from 8.07% to 27.51%, a change of approximately 1.24 standard deviations.

The estimated interpretation response is 26 allocation basis points per standard deviation. The implied additional reallocation is

$$\widehat{a}^I = \frac{25.5665 \times 1.23782}{10{,}000} \simeq 0.003165. \tag{J.16}$$

Thus, stronger interpretation increases the ESG allocation share by 32 basis points, or approximately 32 cents per 100 dollars of initial capital, over the event quarter and the following quarter. The 95% confidence interval is 7–56 basis points.

5.3. Aggregate demand and the stock-price response I consider three scenarios, depending on which investors are activated: ETF investors, mutual-fund and ETF investors together, and institutional equity investors. The institutional scenario is the main counterfactual. Table J.1 collects the calculations using the formulas in Parts 1–4, with full activation ($\alpha = 1$) within each investor base and the additional reallocation applied once to each investor's initial capital.

Table J.1: Additional ESG demand and stock-price responses by activated investor base

	ETFs	Mutual funds + ETFs	Institutional equity investors
Additional ESG investment T^I (\$bn)	19.91	61.51	—
Proportional ESG flow f^I (%)	30.47	26.95	—
Reallocation X^I (\$bn)	20.12	62.24	79.21
Relative stock flow $f^{\text{rel},I}$ (bp)	7.73	23.90	30.41
Relative stock return (bp)	8.62	26.65	33.92
Relative valuation change (\$bn)	14.43	44.64	56.81

All scenarios use the December 2022 multipliers from Part 3. The fund-flow measures T^I and f^I apply to the first two scenarios; stock flows, returns and valuation changes compare green with nongreen stocks. For the institutional scenario, the 95% confidence intervals are 7–60 basis points for the return difference and \$12.5–\$101.1 billion for the valuation difference. These reflect uncertainty in the estimated local response, holding capital and calibration assumptions fixed.

In November 2024, US domestic-equity ETFs managed approximately $K = \$6.29$ trillion, including $G_0 = \$65.35$ billion in ESG ETFs, corresponding to an initial ESG share of $s \simeq 1.039\%$. The first scenario applies the estimated interpretation increment to this capital.

The evidence that mutual-fund flows respond to climate interpretation supports extending the calculation from ETFs to mutual funds and ETFs together. I assume that investors in both types of fund make the same additional ESG allocation per dollar of capital as the investors represented in the local estimate.

In November 2024, US domestic-equity mutual funds and ETFs managed approximately $K = \$19.44$ trillion, including $G_0 = \$228.20$ billion in ESG funds, corresponding to an initial ESG share of $s \simeq 1.174\%$.

The evidence on mutual-fund managers' rebalancing in response to interpretation motivates extending the exercise to institutional investors' stock demand. In this main counterfactual, I assume that institutional investors make the same additional ESG-versus-benchmark reallocation per dollar of capital. The fund-sector calculation implies a reallocation rate of

$$\frac{X^I}{K} = \frac{\$62.24 \text{ billion}}{\$19.44 \text{ trillion}} \simeq 0.320\%. \tag{J.17}$$

In December 2022, institutions held $K_{\text{inst}} = \$24.74$ trillion of the stocks covered by the calibration, or 73.8% of their total market value. Applying the same reallocation rate gives

$$X^I_{\text{inst}} = K_{\text{inst}} \frac{X^I}{K} \simeq \$79.21 \text{ billion}. \tag{J.18}$$

Institutional holdings include stocks held by fund managers on behalf of their investors, so the underlying equity capital is counted once. The price multipliers retain accommodation by all investors, with the additional demand generated by institutions.

As a broader counterfactual, I extend the same reallocation rate to all investors in the covered equity market. Applying it to the \$33.53 trillion of equity capital covered by the December 2022 calibration, $K_{\text{all}} = V_G + V_N$, gives

$$X^I_{\text{all}} = K_{\text{all}} \frac{X^I}{K} \simeq \$107.36 \text{ billion}. \tag{J.19}$$

This implies 46 basis points of green-minus-nongreen appreciation and \$76.99 billion in relative valuation

change. This broader scenario includes institutional and non-13F owners, counting the underlying equity capital once.

5.4. Comparison with annual ESG returns Consider a year in which stronger climate interpretation induces additional reallocations equivalent to every institutional investor making the estimated adjustment once, at unchanged attention. These adjustments are completed during the year, holding the national information environment and other conditions unchanged.[64]

In dollar terms, this is the difference in annual ESG-versus-benchmark demand between the stronger- and weaker-interpretation scenarios:

$$\underbrace{X_{\text{year}}^{\text{stronger } I} - X_{\text{year}}^{\text{weaker } I}}_{X_{\text{inst}}^{I}} \simeq \$79.21 \text{ billion.} \tag{J.20}$$

Applying the calibrated price response gives 34 basis points more green-minus-nongreen appreciation in the stronger-interpretation scenario.

In van der Beck's (2026) decomposition, the average annual return spread is 223 basis points, with 186 basis points attributed to institutional ESG demand. Table J.2 expresses each calibrated effect as a percentage of these benchmarks.

Table J.2: Calibrated appreciation relative to annual ESG-return benchmarks

Activated investor base	Percent of observed annual ESG return spread	Percent of annual institutional ESG-demand contribution
ETFs	3.9	4.6
Mutual funds + ETFs	11.9	14.3
Institutional equity investors	15.2	18.2

This additional appreciation is equivalent in magnitude to 18.2% of the annual contribution attributed to institutional ESG demand and 15.2% of the observed annual return spread.[65]

For the broader all-equity counterfactual, the corresponding ESG-demand benchmark includes both institutional and non-13F demand and equals 155 basis points annually. The calibrated 46-basis-point effect is equivalent to 29.7% of that contribution and 20.6% of the observed annual spread.

[64] I measure the allocation adjustment from the quarter preceding the disaster through the following quarter, allowing for events occurring late in the quarter and for delayed portfolio adjustment. In the annual counterfactual, each investor's additional adjustment is counted once.

[65] These comparisons measure the size of the calibrated effect. Estimating interpretation's contribution to historical returns would require actual activation and interpretation changes over that sample. The calibration uses December 2022 portfolio weights and price responses; the historical returns reflect changing portfolios and market conditions and include dividends.